

Integrated Framework for Surgery Scheduling Management under Uncertainty: From Strategic to Tactical Planning

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Abstract

Abstract text.

Keywords:

1. Introduction

2. Related Works

Surgery scheduling problems have traditionally been classified into three planning levels with different sources of uncertainty: strategic, tactical, and operational (Rahimi and Gandomi, 2020; Cardoen et al., 2010; Wang et al., 2021; Melo and Fogliatto, 2025). At the strategic level, long-term decisions define the desired split of the capacity into specific surgical specialties and specify patient volumes of different medical specialties to be admitted over the planning horizon (Burdett et al., 2023) — a process commonly referred to as Case Mix Planning (CMP). Different objectives have been introduced at this level, such as maximizing profit (Fügner, 2015; Ma and Demeulemeester, 2013) or minimizing long-term costs (Hof et al., 2015; McRae and Brunner, 2020). The tactical level defines a cyclic schedule (often weekly) commonly known as Master Surgery Schedule (MSS) (Bovim et al., 2020; Siqueira et al., 2018; Carneiro et al., 2025). The MSS step allocates time blocks in operating theatres to surgical teams and is reviewed in a medium-term horizon (e.g., monthly or quarterly). Finally, the operational level focuses on the daily operation and derives a Surgical Case Schedule (SCS), which generally assigns individual patients to time blocks previously allocated to specific surgical teams at the tactical level (Vieira et al., 2025; Harris and Claudio, 2022).

Despite the large body of literature in surgery scheduling, there are significant gaps to be bridged. For example, inter-level integration remains limited in scope, often focusing on short-term alignment between surgeries and post-surgical bed occupation, and generally relying on deterministic approaches (Melo and Fogliatto, 2025; Wang et al., 2021; Siqueira et al., 2018). Furthermore, regardless of the planning level, strategies with a focus on guaranteeing waiting time requirements for patients still remain to be proposed. This is probably due to the complex propagation of uncertainty over different planning levels, which demands hybrid approaches that combine mathematical programming, stochastic modelling and queuing theory.

The inherent interconnection between the strategic and tactical levels has motivated unified frameworks that link CMP and MSS decisions (Melo and Fogliatto, 2025). These often adopt sequential or joint optimisation approaches where case mix decisions act as the input to surgery scheduling models (Ma and Demeulemeester, 2013; Fügner, 2015). Modelling approaches typically consist of Mixed Integer Programming (MIP) formulations that focus on maximising hospital profit subject to Operating Theatre (OT) and bed capacity constraints. While these frameworks acknowledge uncertainty, it is generally confined to downstream elements such as length of stay or ward occupancy.

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Although modelling post-surgical congestion is relevant, such partial treatments remain short-sighted when not complemented by uncertainty in surgery durations and, crucially, by an explicit modelling of dynamic waiting lists across surgical specialities, which constitute the starting point of the entire planning process.

The perspective can be broadened by incorporating additional uncertainty sources and dynamic elements. For instance, Liu et al. (2019) model elective admissions through a Markov Decision Process (MDP), highlighting the role of waiting-list size in guiding strategic decisions. Despite its conceptual value, the framework disregards essential practical constraints, such as surgeon availability, block scheduling and case-mix heterogeneity, thereby limiting practical applicability. Similarly, McRae and Brunner (2020) demonstrate that explicitly modelling uncertainty and selecting appropriate aggregation levels can significantly improve CMP outcomes and enable joint CMP-MSS optimisation. Nevertheless, waiting-list dynamics remain outside of the modelling scope, implying no full integration of strategic and tactical decisions. Furthermore, their reliance on Sample Average Approximation (SAA) leads to representations that may understate distributional variability. These studies underscore the value of integrating CMP and MSS, yet they also reveal persistent shortcomings: uncertainty is represented only selectively across planning levels, waiting-list requirements are rarely modelled explicitly, and key operational constraints required for real-world implementation remain overlooked.

Uncertainty in surgery duration is another important issue that, when disregarded, may result in unpredictable idle or overtime in OTs (Shehadeh and Padman, 2022). Whilst overtime is costly and can lead to cancellations that compromise the quality of care, idle time reflects an inadequate use of OT capacity (Shehadeh and Padman, 2022; Harris and Claudio, 2022; Kayvanfar et al., 2025). With the aim of maximising hospital revenue while preventing excessive overtime, Barz and Rajaram (2015) propose a tactical patient admission and scheduling problem that derives a CMP under multiple resource constraints and stochastic clinical progression. The approach utilises a newsvendor heuristic to reserve capacity for emergency arrivals, and an MDP model to select the patient cohort to be admitted whilst considering probabilistic transitions between clinical states. The model, however, excludes the possibility of cancellations or postponements driven by overtime, as it assumes that the hospital always disposes of the resources to treat each admitted patient. Moreover, by treating resource consumption as deterministic, the framework effectively ignores uncertainty in surgery duration — one of the main drivers of operational instability. As a result, the model underestimates the practical interplay between stochastic surgery durations, cancellations and overtime, limiting its applicability to integrated CMP-MSS decision-making.

Furthermore, Zhang et al. (2020) study a tactical surgery planning with the explicit aim of mitigating overtime risk when allocating elective patients to OTs over an upcoming planning horizon. Their model accounts for stochastic surgery durations and adopts a risk-averse objective that jointly captures both the probability and the expected magnitude of overtime, extending earlier approaches that consider only one of these dimensions (Shylo et al., 2013; Rath et al., 2017). Although they derive the durations from real-world data, eventually they approximate it through a set of discrete scenarios, limiting the full distributional representation. A similar treatment appears in Shehadeh et al. (2026), which introduces a Distributionally Robust Optimisation (DRO) approach to address uncertainty in surgery durations. While their approach minimises worst-case expected cost over an ambiguity set defined by known moments and support, it likewise relies on scenario-based representations. Accurately capturing surgery-duration variability requires a large number of scenarios, substantially increasing computational burden and potentially undermining the practical advantages of DRO and related methods.

These limitations regarding the integration of overtime management and its associated cancellation risks into surgical planning frameworks highlight important research gaps. There remains a need for models that explicitly incorporate cancellation dynamics induced by overtime and that provide a more thorough representation of surgery-duration uncertainty beyond scenario-based approximations. A key challenge which we tackle in this paper is to develop a modular approach to capture full distributional characteristics to build a planning model and use its output to feed an integrated CMP-MSS approach — i.e., not solving an entire model at once.

An additional and often underexplored dimension in integrated surgical planning concerns the incorporation of practical constraints inherent to real-world healthcare settings. Certain surgical activities, such as organ transplantation, impose rigid scheduling requirements that are difficult to reconcile with standard optimisation assumptions (da Silva et al., 2025). Furthermore, public hospitals in developing countries (e.g., Brazil) operate under severe resource constraints, high and volatile demand, and heterogeneous case mixes, all of which exacerbate the complexity of surgical planning (Carneiro et al., 2025; Siqueira et al., 2018; da Silva et al., 2025). These contextual factors challenge the scalability and transferability of many optimisation-based approaches, which often presume stable resources,

predictable demand and homogeneous patient populations. Indeed, as noted by Melo and Fogliatto (2025), much of the existing literature is calibrated to relatively small hospitals in developed regions, limiting its relevance for large, resource-constrained systems. Ignoring such operational realities risks producing models that are theoretically stable but difficult to implement in practice, underscoring the need for scheduling frameworks that explicitly accommodate institutional, organisational and contextual constraints.

This paper addresses the identified gaps in the surgical planning literature by proposing an integrated framework that jointly links strategic and tactical decision levels while explicitly accounting for key sources of uncertainty. The framework captures dynamic surgical demand through explicit waiting-list modelling, enabling the enforcement of waiting-time requirements and long-term queue stability (Shortle et al., 2018), and incorporates stochastic surgery durations to properly balance cancellation, idle time, and overtime risks for each surgical session. The proposed approach is grounded in a real-world application at the Clinical Hospital of the University of Campinas (HC-Unicamp) in Brazil, a high-complexity public hospital that provides detailed operational data and a challenging planning environment (HC - Unicamp, 2024b).

The main contributions of this work are as follows:

- A stochastic waiting-list management strategy based on an (R, Q) control policy (Axsäter, 2015) that ensures compliance with waiting-time targets while maintaining long-term queue stability across surgical specialties;
- A cancellation risk management mechanism based on a newsvendor formulation (Arrow et al., 1951), which determines optimal time buffers for the last surgery in each session under duration uncertainty and defines probability distributions for the number of planned surgeries that can be accommodated without incurring overtime;
- A MIP model for operating theatre allocation that: (i) incorporates practical constraints, including those associated with transplant surgeries, while remaining computationally tractable; and (ii) explicitly incorporates the waiting-list management strategy and cancellation risk considerations from the previous items;
- An empirical validation of the integrated framework with data from a large, resource-constrained public hospital, demonstrating its operational relevance and implementability.

By jointly addressing relevant sources of uncertainty within a unified framework, this work introduces an innovative and integrated surgical planning approach that advances the literature by bridging multiple relevant gaps. The approach also bridges the gap between theory and practice by bringing forward a practically grounded and empirically validated framework for long-term management of surgery queues within complex healthcare settings.

3. Description of the case study

Hospital de Clínicas Unicamp (HC - Unicamp) is a university hospital located in Campinas, Brazil (HC - Unicamp, 2024b). It is a high-complexity public hospital serving approximately 6.5 million inhabitants from Campinas' metropolitan area and other regions of Brazil that handles around 500,000 patient cases per year. Of these, roughly 10,000 correspond to elective surgical procedures, with the remainder involving outpatient consultations and emergency care (HC - Unicamp, 2024a). In addition to providing healthcare services, HC - Unicamp operates as a teaching and research hospital, training healthcare professionals and conducting medical and interdisciplinary research.

Elective surgical patients follow a complex, multi-stage care pathway (see Figure 1), encompassing pre-operative, intra-operative and post-operative phases. This process begins with waiting-list management, initiated at local medical consultations and coordinated through the regional bed regulation centre (HC - Unicamp, 2024c), a government agency responsible for regulating hospital access across Campinas Metropolitan Region. Patient entry into HC - Unicamp typically occurs via scheduled outpatient appointments, although some emergency cases subsequently become elective and are incorporated into the same planning process. Once referred, patients are assigned to waiting lists according to medical subspecialty. Decisions at this stage directly determine the CMP (i.e., strategic layer), influencing waiting times, speciality-level demand balance and the long-term utilisation of operating theatre capacity (Siqueira et al., 2018).

On the day of surgery, patients enter the upstream unit management phase, where clinical reassessment is performed. Based on clinical requirements, Intensive Care Unit (ICU) support may be required pre- or postoperatively,

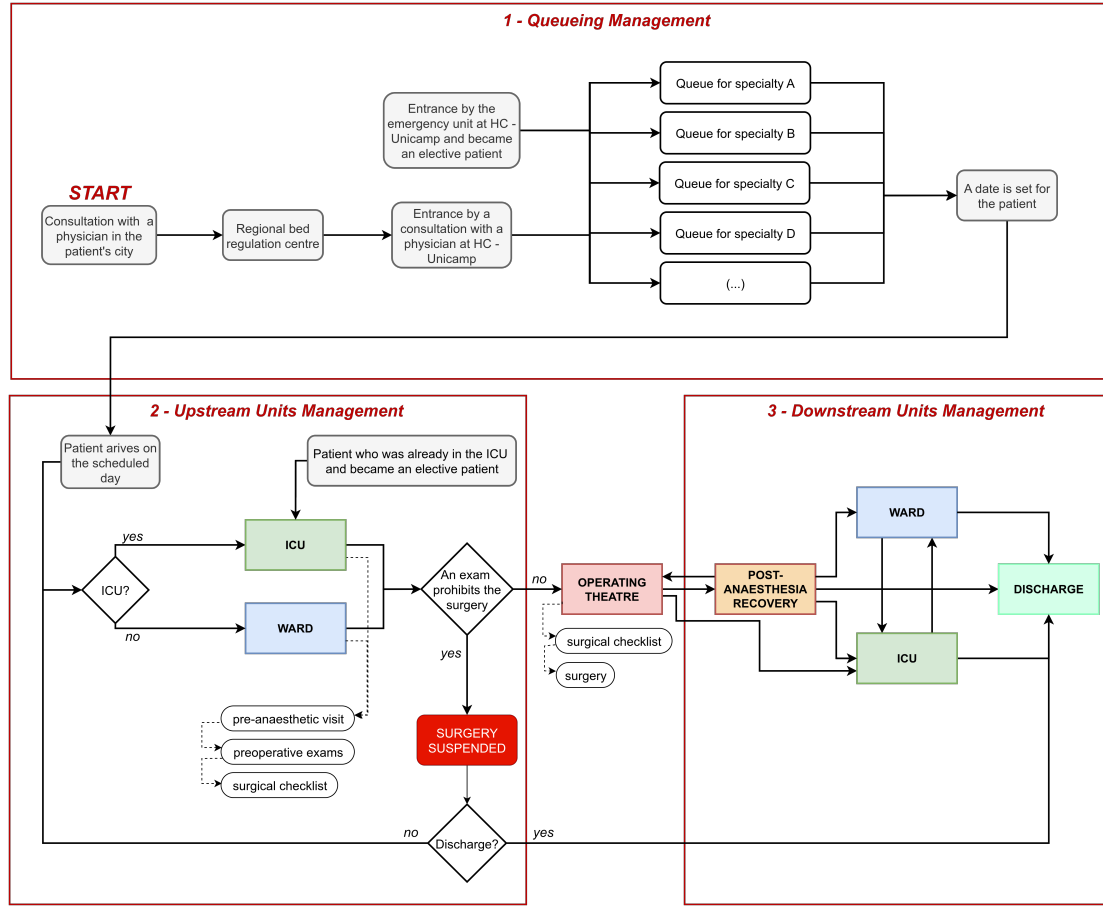


Figure 1: Patient flow at HC - Unicamp.

necessitating prior bed reservation; otherwise, postoperative ward bed availability must be ensured. Patients then undergo pre-anaesthetic evaluation, diagnostic tests and surgical safety checks. If contraindications arise, the procedure is suspended and the patient is either discharged or rescheduled, according to clinical severity. If cleared, the patient proceeds to the operating theatre, where the surgical procedure and subsequent theatre cleaning are performed. In the present study, ward and ICU capacity are sufficient and therefore do not constitute binding constraints, allowing the analysis to focus on waiting-list management, uncertainty in surgery duration and operating theatre allocation.

Once in the operating theatre, the patient is managed by a dedicated surgical team and anaesthetist. This stage includes final safety checks, the surgical procedure itself and post-operative cleaning. Here, uncertainty in surgery duration plays a central role, as it directly affects idle time, overtime, and the risk of cancellations. Indeed, it is well known that surgery duration variability has a significant impact on operating theatre utilisation and overall system throughput (e.g., Pandit and Tavare, 2011), therefore meriting explicit consideration in the proposed framework.

Following surgery, patients are transferred to the Post-Anaesthesia Recovery Unit (PARU), with direct ICU transfer occurring only in exceptional cases. After stabilisation, patients are admitted to a ward or ICU bed and subsequently discharged. HC - Unicamp has 15 operating theatres dedicated to elective surgeries: 13 located in the Elective Surgery Centre (ESC) and 2 in the Ambulatory Surgery Centre (ASC), which operates exclusively during morning shifts. These theatres are heterogeneous with respect to infrastructure and equipment, leading to compatibility restrictions across medical subspecialties.

The planning horizon comprises approximately 20 working days per month, each divided into two 6-hour shifts (morning and afternoon). Henceforth, we refer to a theatre shift as a *block*. Hospital policy assigns each block to a

single subspecialty, while allowing the same subspecialty to occupy multiple blocks within the same theatre, provided compatibility conditions are met. Sequential allocation is therefore permitted (and preferred by hospital managers), enabling a subspecialty assigned to a morning block to also occupy the corresponding afternoon block.

Each month, operating theatres must be allocated across 40 medical subspecialties. Relevant subsets include: (i) those that mandatorily require two consecutive blocks in the same theatre due to the extended duration of their procedures; (ii) priority subspecialties, which take precedence in allocation decisions in accordance with institutional rules; and (iii) transplant subspecialties, which impose additional operational constraints.

Block allocation feasibility depends mainly on the availability of surgical teams and anaesthetists. A subspecialty can only be assigned to a block if an appropriate team is available on the corresponding day and shift. Moreover, the number of simultaneous allocations requiring anaesthesia cannot exceed the number of anaesthetists available in each period. For example, if 15 anaesthetists are available on a given morning, at most 15 anaesthesia-dependent procedures may be scheduled concurrently.

Day	PERIOD/THEATRE	1	2	3	4	5	6	7	8	9	10	11	12	17	ASC-1	ASC-2
1	MORNING	card sur	tumour orth		procto	uro onco	anaest dor		gast liv	otorhi			vasc			
	AFTERNOON	card sur	tumour orth	pc local	procto	uro onco				otorhi	gast gdp/vb		vasc	vasc local		
2	MORNING	card sur	foot orth	plast	trauma orth	uro tx		uro tx	gast eed	otorhi	gast eed	neuro	trauma orth			
	AFTERNOON	card sur		plast						otorhi	gast eed	neuro	trauma orth			
3	MORNING	card sur	knee orth	plast	thorax sur	vasc local	paed sur	plast		otorhi	head neck	neuro	trauma orth	vasc		
	AFTERNOON	card sur		plast	thorax sur		paed sur	plast		otorhi	gast obes	neuro		vasc		
4	MORNING	card sur	eef	adult endo	gast gdp/vb	uro lithiasi	vasc	adult endo	gast liv	otorhi	gast obes	neuro	neuro colu		ophtal	
	AFTERNOON	card sur	pc local	ophtal	gast gdp/vb	uro lithiasi	vasc	otorhi	gast liv	otorhi	gast obes	neuro	neuro colu			
5	MORNING		paed orth	plast		uro gyneco	paed sur			head neck	head neck	neuro colu			hand orth	
	AFTERNOON		hip orth										trauma orth			

Figure 2: Excerpt from a week's operating theatre schedule.

Figure 2 presents an excerpt from a monthly operating theatre schedule. Rows correspond to day and shift and columns to theatres; allocated blocks are colour-coded (blue for procedures requiring anaesthesia and grey otherwise). Unallocated (blank) slots reflect the complexity of the constraint set. Allocations shown in red correspond to transplant surgeries, which require the simultaneous allocation of two theatres within the same day and shift.

Historically, schedules such as those shown in Figure 2 have been manually constructed using only electronic spreadsheets, often requiring more than two full working days. This manual process is time-consuming and does not guarantee optimality with respect to institutional policies. Automating the scheduling process therefore reduces administrative burden and enables the generation of optimal solutions under complex constraints.

In addition to the impact of automating surgery scheduling in a reference hospital in Brazil, this paper provides an approach that is firmly grounded in reality and is therefore adaptable to other general hospitals. By jointly managing waiting-list dynamics, surgery duration uncertainty and operating theatre allocation, our proposed approach provides a coherent and practically grounded solution capable of satisfying waiting-time requirements while mitigating idle time, overtime and cancellation risks.

4. Proposed Framework

We propose an integrated framework for elective surgery planning under uncertainty that links waiting-list management, in-block time management, and operating theatre scheduling. Rather than formulating these decisions as a single monolithic model, the framework decomposes the problem into three inter-related modules. The first two modules address strategic decisions related to queue and overtime management, and their outputs are then used as inputs to the tactical operating theatre scheduling model. Figure 3 summarises the proposed framework.

The queue management module establishes targets for the number of surgeries to be performed at each planning period (e.g., monthly). Specifically, a modified (R, Q) policy defines, for each medical sub-specialty, a lower and

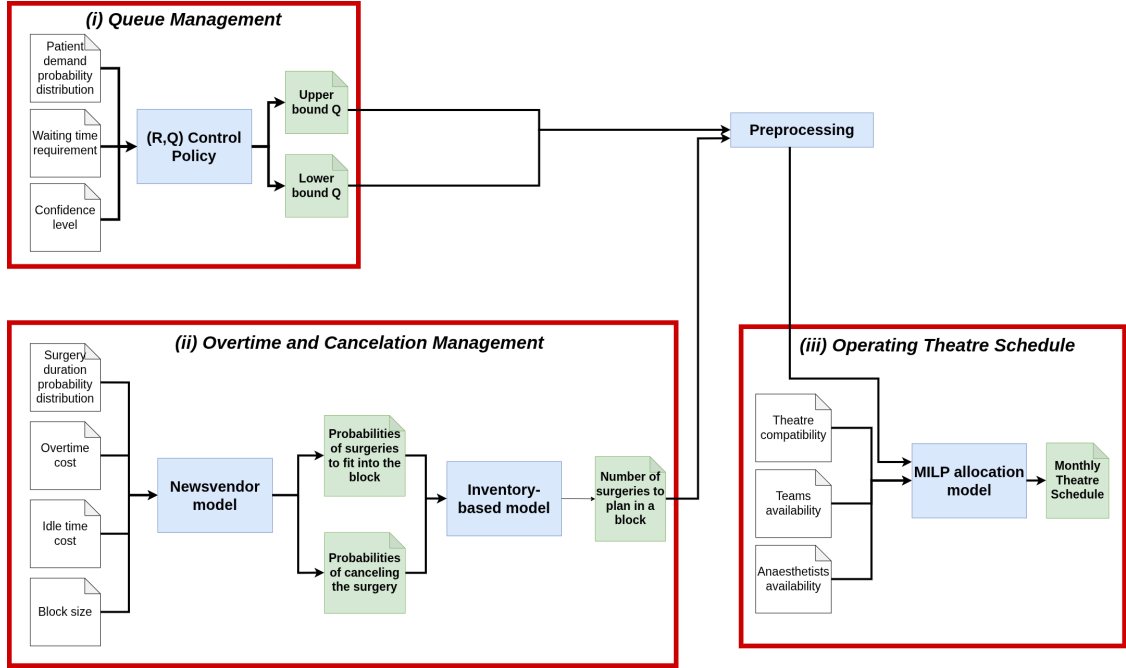


Figure 3: Proposed framework.

an upper bound on the number of patients to be admitted for surgery during the planning period, while ensuring that system-wide waiting-time requirements are satisfied in spite of the uncertainty in patient arrivals.

The second module balances overtime, idle time and the risk of surgery cancellation due to overbooking. Considering the uncertainty in surgery duration, a Newsvendor model guides dynamic decisions on whether to perform or postpone the next surgery, given the time remaining in the session. The output of the Newsvendor model then informs an inventory model that determines the number of surgeries to be booked in a block while balancing the expected number of surgeries with the risk of cancellation.

The outputs of the first two modules are then combined to provide key inputs to the third module, which formulates the OT scheduling problem. Specifically, given the probability distribution of the number of performed surgeries per session (module 2), and the bounds on the number of surgeries to be performed in the period (module 1), one can derive lower and upper limits on the number of blocks to be allocated to each surgical sub-speciality. Hence, theatre allocation decisions are informed not only by resource availability and clinical compatibility, but also include constraints to guarantee that waiting-time requirements can be met despite the uncertainties in the demand and duration of surgeries. Crucially, the scheduling model remains deterministic, thereby facilitating practical implementation and ensuring rapid solution times.

Overall, the proposed framework provides a coherent approach to elective surgery planning across multiple decision levels. By explicitly incorporating uncertainty into the strategic planning stages, it supports more robust and operationally meaningful scheduling decisions that are guaranteed to align to strategic waiting-time requirements. The following subsections present each module in detail. To help navigate the paper, Table 1 consolidates the list of acronyms.

4.1. Module 1: Long-Term Waiting List Management

CMP is the strategic decision layer that decides the volumes of elective surgeries to be performed over a given planning horizon, see Section 2. In contrast to much of the existing literature, which primarily selects these volumes to optimise financial or capacity-related objectives, this work adopts a stability-oriented perspective. We aim to enable decision-makers to impose maximum acceptable waiting times (e.g., six months) and to guarantee, in the long run, that such targets are met without allowing waiting lists to grow unboundedly. This is particularly relevant in the

Table 1: Acronyms used in the paper.

Acronym	Description
CMP	Case Mix Planning
MSS	Master Surgery Scheduling
SCS	Surgical Case Scheduling
OT	Operating Theatre
MDP	Markov Decision Process
SUS	Brazilian Unified Health System
HC - Unicamp	Hospital de Clínicas Unicamp
OTSP	Operating Theatre Scheduling Problem
MILP	Mixed-Integer Linear Programming
PMF	Probability Mass Function
CDF	Cumulative Distribution Function

context of public healthcare systems, such as Brazil’s Unified Health System (SUS), where there is a legal obligation to provide care to all patients in need. Thus, ensuring equitable access to care and adherence to waiting time standards is paramount.

In line with this objective, this section introduces the theoretical foundations of the modified (R, Q) control policy proposed to manage elective surgery waiting lists under stochastic demand. We first characterise the dynamics of the underlying stochastic process and analyse its long-run behaviour. Subsequently, we derive analytical upper and lower bounds on the batch size Q that ensures compliance with prescribed waiting-time requirements, thereby establishing the strategic input for the integrated planning framework. Table 2 summarises the notation used in this section.

4.1.1. (R, Q) surgery scheduling policy: long-term distributions

Let X_t , $t \geq 0$ be a stochastic process representing an elective surgery queue with stationary and independent increments over discrete planning periods, with X_t representing the number of patients in the system at period $t \geq 0$. Assume that the queue is controlled via a modified (R, Q) policy (e.g., Axsäter, 2015) whereby a batch of $Q \in \mathbb{N}$ surgeries is processed at time period t whenever $X_t \geq R$, $R \geq Q$ at the start of the period, where $\mathbb{N}$ is the set of positive integers. In contrast to the classical (R, Q) policy, which processes as many batches as necessary to bring the inventory below the reorder point (R), our policy can only process a batch at a time and relies on long-term properties to ensure that the process X_t , $t \geq 0$ is stable and guarantees prescribed waiting time requirements.

Let D be a non-negative integer random variable representing the overall demand between two consecutive planning periods, with probability distribution $f_D : \Omega \rightarrow [0, 1]$, $\Omega \subset \mathbb{Z}_+$, and cumulative distribution $F_D : \mathbb{Z}_+ \rightarrow [0, 1]$ given by

$$F_D(d) = \sum_{l=0}^d f_D(l),$$

where $\mathbb{Z}_+$ denotes the set of non-negative integers. This implies that the dynamics of process X_t , $t \geq 0$ can be expressed as:

$$X_{t+1} = X_t + d - Q \mathbb{1}_{\{X_t \geq R\}}, \quad (1)$$

where $\mathbb{1}_{\{A\}}$ is the indicator function, which is equal to 1 if A holds and is nil otherwise, and d is a value drawn from f_D that represents the surgical demand in period t . The state space of process X_t , $t \geq 0$ is given by $S = \{0, 1, 2, \dots\}$, where $X_t = i$ indicates that there are $R - Q + i$ patients waiting for elective surgery.

Letting

$$Y_t = \left\lfloor \frac{X_t}{Q} \right\rfloor, \quad Z_t = X_t \% Q, \quad (2)$$

where $\%$ represents the modulus operator, we can write:

$$X_t = QY_t + Z_t,$$

where Y_t , $t \geq 0$ is the integer number of batches required to bring the queue instantaneously below R and $Z_t \in \{0, 1, \dots, Q - 1\}$ is the number of patients that would remain in excess of the minimum allowed queue of $(R - Q)$ patients after removing Y_t batches of Q patients instantaneously.

Table 2: Symbols used in the long-term waiting list management model.

Index	Description
t	Planning period index
z	State index of the remainder process
i, j	Generic state indices
Set	Description
S	State space of process X_t
S_Y	State space of process Y_t
S_Z	State space of process Z_t
Ω	Support of the demand random variable
Parameter	Description
R	Reorder threshold of the modified (R, Q) policy
Q	Batch size under the modified (R, Q) policy
τ	Waiting-time target
α	Probability threshold for the waiting-time requirement
$\bar{Q}$	Upper bound on feasible batch sizes
$\underline{Q}$	Lower bound on feasible batch sizes
M	Maximum number of batch arrivals in one period
Random Variable / Stochastic Process	Description
X_t	Queue-length process
D	Demand between two consecutive planning periods
d	Realisation of D
Y_t	Batch-level queue process
Z_t	Remainder process associated with X_t
A	Number of batch arrivals in one period
ω	Waiting time of an arriving patient
Function	Description
$f_D(\cdot)$	Probability mass function of D
$F_D(\cdot)$	Cumulative distribution function of D
$A(z)$	Number of batch arrivals given remainder state z
$\pi_Z(\cdot)$	Steady-state distribution of Z_t
$\pi_Y(\cdot)$	Steady-state distribution of Y_t
P_Z	Transition matrix of process Z_t
P^Y	Transition matrix of process Y_t
$p_{zz'}^Z(k)$	Conditional transition probability of Z_t given demand k
p_{ij}	Generic transition probability entry
Derived Quantity	Description
$\mathbb{1}_{\{X_t \geq R\}}$	Indicator of batch processing

Note that, from the definitions above, the state space of process Y_t , $t \geq 0$ is $S_Y = \mathbb{Z}_+$, while $S_Z = \{0, 1, \dots, Q-1\}$ is the state space of process Z_t , $t \geq 0$. In the next result, we will derive the long-term distribution of process Z_t , $t \geq 0$.

Proposition 1. *Let $\pi_Z : S_Z \rightarrow [0, 1]$ be the steady state distribution of process X_t , $t \geq 0$. Then, it holds that*

$$\pi_Z(z) = \frac{1}{Q}, \forall z \in S_Z. \quad (3)$$

Proof. The proof closely follows that of Axsäter (2015, Proposition 5.1). Let P_Z be the transition matrix describing the dynamics of process Z_t , $t \geq 0$. Assume that $Z_t = z$, $z \in S_Z$ at the start of period $t \geq 0$, and let $D = k$ be the number of new patient arrivals in period t . Hence, it follows that

$$Z_{t+1} = (z + k) \% Q = (z + k + QY_t) \% Q = (X_t + k) \% Q,$$

where the last equality come from the definitions in Eq. (2). Therefore, given $D = k$, it is evident that $p_{zz'}^Z(k) := P(Z_{t+1} = z' | Z_t = z, D = k)$ equals one for exactly one value of z and zero otherwise. Consequently, we can write:

$$\sum_{z=0}^{Q-1} p_{zz'}^Z = \sum_{z=0}^{Q-1} \mathbb{E}_k \{p_{zz'}^Z(k)\} = \mathbb{E}_k \left\{ \sum_{z=0}^{Q-1} p_{zz'}^Z(k) \right\} = \mathbb{E}_k \{1\} = 1.$$

The expression above implies that the Markov chain describing the dynamics of process Z_t , $t \geq 0$ is doubly stochastic, and therefore has a uniform steady state distribution. Indeed, substituting (3) in the steady state equations:

$$\pi^Z(z) = \sum_{z'=0}^{Q-1} \pi^Z(z') p_{z'z}^Z,$$

one can easily see that the steady state distribution satisfies Eq. (3). $\square$

The result in Proposition 1 is very important, as it shows that the steady state distribution of process Z_t , $t \geq 0$, is uniform regardless of the distribution of patient arrivals. Next, we will use this fact to derive the steady state distribution of process Y_t , $t \geq 0$.

4.1.2. The dynamics of process Y_t , $t \geq 0$

Assume that process Z_t , $t \geq 0$ has reached equilibrium, which is also equivalent to supposing that $P(Z_0 = z) = \pi_Z(z) = \frac{1}{Q}$ (e.g., Brémaud, 2020). We can describe the dynamics of process Y_t , $t \geq 0$ as:

$$Y_{t+1} = \begin{cases} Y_t - 1 + A, & \text{if } Y_t \geq 0, \\ A; & \text{if } Y_t = 0, \end{cases} \quad (4)$$

where A is the round number of batch arrivals at time t . To fully characterize the random variable A , it is necessary to define

$$A(z) = \left\lfloor \frac{D+z}{Q} \right\rfloor, \quad (5)$$

which represents the round number of batch arrivals at time t , given that $Z_t = z$. Then, in view of Proposition 1, it follows that:

$$P(A = k) = \frac{1}{Q} \sum_{z=0}^{Q-1} P(A(z) = k). \quad (6)$$

Therefore, from (4) we have:

$$p_{ij}^Y = P(Y_{t+1} = j | Y_t = i) = \begin{cases} P(A = j), & \text{if } Y_t = 0, \\ P(A = j - i + 1), & \text{if } Y_t \geq 1. \end{cases} \quad (7)$$

Note that Eq. (4)-(7) correspond to the dynamics in Shortle et al. (2018, Section 6.1.2). Similarly to Shortle et al. (2018), the transition matrix of process Y_t , $t \geq 0$ can be written as:

$$P^Y = [p_{ij}^Y] = \begin{bmatrix} k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ 0 & k_0 & k_1 & k_2 & \dots & k_M & 0 & \dots \\ 0 & 0 & k_0 & k_1 & k_2 & \dots & k_M & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \quad (8)$$

where M is the maximum number of new batch arrivals, i.e., the maximum value that random variable A can assume.

The steady state probabilities of process Y_t , $t \geq 0$ can then be recursively evaluated by solving:

$$\pi_Y^T = \pi_Y^T P, \quad (9)$$

with the contour condition that $\pi_Y(0) = 1 - \rho$ (Shortle et al., 2018, Eq. (6.12)), where $\rho = E(A)$. Observe that, to guarantee stability, we must have $E(A) < 1$. Note also that the steady state probabilities up to a given state j can be recursively calculated using the first j equations in (9).

4.1.3. Upper and lower bounds on the batch size Q under waiting time requirements

Let ω be the waiting time of a patient joining the queue, and assume that the healthcare system requires that:

$$P(\omega > \tau) < \alpha, \quad (10)$$

for some target waiting time $\tau > 0$ and probability threshold $\alpha \in (0, 1)$. Now, if a patient arrives and encounters $Y_t > 0$ batches of patients in the queue, their turn will only arrive once these Y_t batches are processed, which will take Y_t time periods. This means that the patient will have to wait $\tau = Y_t$ periods for their turn. Therefore, Eq. (10) can be equivalently stated as:

$$\sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha, \quad (11)$$

where $\pi_Y : S_Y \rightarrow (0, 1)$ satisfies Eq. (9). Therefore, to limit the waiting time according to the prescribed requirements, we can establish a lower bound ($\underline{Q}$) on the batch size $Q > 0$ as follows:

$$\underline{Q} = \min\{Q \in \mathbb{N} : \sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha\}. \quad (12)$$

When $Y_t = 0$, on the other hand, the next batch will only be processed when the system leaves this state, therefore by applying the strong Markov property (Brémaud, 2020), we have:

$$\omega = \min\{\delta > 0 : Y_\delta > 0 | Y_0 = 0\}.$$

Considering the transition matrix in Eq. (7), it follows that ω will be a geometrically distributed variable and:

$$P(\omega > \tau | Y_0 = 0) = k_0^\tau. \quad (13)$$

One can intuitively see that the value of k_0 will increase as Q increases, since in this case we will need more patient arrivals until we are able to process a full batch of patients. Hence, from Eq. (13), we can set up an upper bound $\bar{Q}$ for the batch size such that:

$$\bar{Q} = \max\{Q \in \mathbb{N} : k_0^\tau < \alpha\}. \quad (14)$$

In the example below, we show how to derive general bounds $\underline{Q}$ and $\bar{Q}$ for a given demand distribution and waiting time requirements.

Example 1 (Deriving lower and upper bounds for Q). *Consider an elective surgery queue with monthly demand D that satisfies the distribution in Table 3.*

Table 3: Monthly demand distribution for surgeries

Demand (d)	0	1	2	3	4	5	6	7	8	9	10
$P(D = d)$	0.0308	0.0393	0.1179	0.2096	0.2445	0.1956	0.1086	0.0414	0.0103	0.0015	0.0005

The expected monthly demand is therefore $\mathbb{E}(D) = 3.9022$. Setting $Q = 4$, which ensures long-term stability, we have $S_Z = \{0, 1, 2, 3\}$. This allows us to evaluate the individual distributions of $A(z)$, $z \in S_Z$ as defined in Eq. (5), which we depict in the Table 4, along with the distribution of variable A from Eq. (6).

The distributions in Table 4 follow directly from Eq. (5) and Eq. (6). For example, when $z = 0$, we know that the queue comprises an integer number of batches. Therefore, $P(A(0) = 0) = p_D(0) + p_D(1) + p_D(2) + p_D(3)$ (Table 3) as up to 3 new arrivals will not suffice to generate a new batch of $Q = 4$ patients. Analogously, $A(0) = 1$ if we observe from 5 to 7 arrivals and $A(0) = 2$ when 8 to 10 patients arrive in the period, as these will suffice for 2 new batches.

Conversely, when $z = 3$, we have three patients in excess of a discrete number of batches, which means that $P(A(3) = 0) = p_D(0)$ as new batches will be formed whenever one or more patients arrive. One new batch will

Table 4: Monthly demand distribution for surgeries

$P(A(z) = d)$	Batch demand (d)			
	0	1	2	3
$z = 0$	0.3976	0.5901	0.0123	0
$z = 1$	0.1880	0.7583	0.0537	0
$z = 2$	0.0701	0.7676	0.1618	0.0005
$z = 3$	0.0308	0.6113	0.3559	0.002
$P(A = d)$	0.171625	0.681825	0.145925	0.000625

form if $D \in [1, 4]$, two new batches if $D \in [5, 8]$ and three new batches when $D \in [8, 10]$. Specifically, when 10 patients arrive, they will form a set of 13 patients when joined to the $z = 3$ remaining patients from the start of the period. These 3 patients will then suffice to form three batches of $Q = 4$ patients, with one remaining batchless for the following period.

Note that $\mathbb{E}(A) = 0.9755$, and that this value is equal to $\rho = \frac{\mathbb{E}(D)}{Q} = \frac{3.9022}{4} = 0.9755$. This is expected, as the dynamics of process Y_t describes the original system, only looking at it in terms of batches of patients. Note that $\mathbb{E}(A)$ is simply the expected number of batches of Q patients arriving and since we process $\mu = 1$ batch per time period, we have $\rho = \mathbb{E}(A)$ when looking at the evolution of the batch process Y_t , $t \geq 0$.

Now, let us suppose that $\tau = 6$ and $\alpha = 0.05$ in (10), which means that the system requires 95% of the patients to be operated in up to six months. To verify whether that is satisfied, we need firstly to solve Eq. (9) for the steady state distribution of Y_t , $t \geq 0$ and then use it to verify (11). From Table 4, we obtain $k_0 = 0.171625$, $k_1 = 0.681825$, $k_2 = 0.145925$ and $k_3 = 0.000625$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.02445$, $\pi_Y(1) = 0.1182$, $\pi_Y(2) = 0.1219$, $\pi_Y(3) = 0.1046$, $\pi_Y(4) = 0.0898$, $\pi_Y(5) = 0.077$, $\pi_Y(6) = 0.0661$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.602 < 0.95 = 1 - \alpha.$$

This implies that $\underline{Q} > 4$ in Eq. (12) and we should, therefore, recalculate the probability above for increasing values of Q until we find the first value that satisfies the waiting time criterion - which will give us the lower bound $\underline{Q}$.

To verify the waiting time requirements when $Y_t = 0$, one can apply (13), which implies that $P(\omega \leq 6 | Y_t = 0) = 0.171625^6 \approx 2.5 \cdot 10^{-5} < 0.05$. Therefore, to find the bound $\underline{Q}$ in Eq. (14), we must iterate on higher values of Q .

When verifying Eq. (12) with $Q = 10$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.609780^6 \approx 0.05114 > 0.05$. Then, if one set $Q = 9$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.566478^6 \approx 0.03304 < 0.05$. Thus, we can set $\underline{Q} = 9$.

For the $\underline{Q}$, one can verify $Q = 5$, which implies $k_0 = 0.265720$, $k_1 = 0.688120$, $k_2 = 0.046160$ and $k_3 = 0$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.21956$, $\pi_Y(1) = 0.606723$, $\pi_Y(2) = 0.143539$, $\pi_Y(3) = 0.024935$, $\pi_Y(4) = 0.004332$, $\pi_Y(5) = 0.000752$, $\pi_Y(6) = 0.000131$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.999973 > 0.95 = 1 - \alpha.$$

Thus, we can set $\underline{Q} = 5$.

4.2. Module 2: Modelling Uncertainty in Surgery Duration and Block Sessions

This section details the components of module 2 of the proposed framework, see Section 4. Firstly, it is necessary to estimate how many elective patients can be accommodated within a single operating theatre block for each sub-speciality, to ensure consistency between the periodic schedule and the waiting-time requirements established by the (R, Q) policy in Section 4.1. This is key to ensuring that the number of sessions allocated to a given speciality is enough to warrant a number of surgeries that lies in the interval $[\underline{Q}, \bar{Q}]$, considering the probabilistic distribution of surgery durations. Specifically, a trade-off arises between idle time, that results in underutilised theatre capacity, and overtime, which may trigger cancellations of scheduled procedures.

At first, we introduce a Newsvendor model to guide operational decisions on whether to perform or cancel the next scheduled surgery, depending on the time remaining in the session. Then, assuming that these decisions are taken sequentially at the end of each scheduled surgery, we introduce an inventory model to find the number of patients that can be reliably scheduled within a time block, as well as the probability distribution of the number of cancellations at each session. These are tactical inputs required for subsequent operating theatre allocation decisions.

4.2.1. Managing surgery cancellations via a Newsvendor-based model

Let us now consider the problem of determining the optimal time budget for the next surgery in a session, based on the random duration of this surgery. The symbols used in this section are defined in Table 5.

Table 5: Symbols used in the Newsvendor-based model.

Symbol	Description
Index	Description
s	Specialty index
Set	Description
$\mathcal{S}$	Set of specialties
Ω	Support of the surgery-duration random variable
Parameter	Description
T	Length of an operating theatre block
t_b	Time budget allocated to the last surgery in a block
c	Unit cost of unused allocated time
p	Unit penalty cost of overtime
t^*	Safety time threshold for starting the next surgery
Random Variable	Description
T_D	Random duration of a surgery
β_1	Unused allocated time
β_2	Overtime duration
Function	Description
$f_{T_D}(\cdot)$	Probability mass function of T_D
$F_{T_D}(\cdot)$	Cumulative distribution function of T_D
$u(t_b)$	Expected cost of allocating time budget t_b

Let T_D be a non-negative random variable taking values from $\Omega \in \mathbb{R}_+$, which represents the duration of a given surgery, with cumulative distribution $F_{T_D} : \Omega \rightarrow [0, 1]$. Assume that, once the theatre is ready for the surgery, it only proceeds as planned if the remaining time in the session exceeds a prescribed budget of $t_b \in \Omega$ time units. Otherwise, the surgery is called off and the session is terminated.

The time budget $t_b \in \Omega$ can be interpreted as a perishable inventory of session time. If the inventory is excessive, we will not fully utilise the time allocated to the surgical team to perform the surgery. Conversely, if we proceed with the surgery but the time budget is insufficient, the surgical team will have to work overtime to complete the surgery. Let $c > 0$ be the cost of each unit of time budget in excess of the actual surgical time T_D and $p > c$ be a penalty for each unit of overtime that the surgical team incurs. The expected cost of the time deviation between T_D and t_b is defined as:

$$u(t_b) = c\mathbb{E}(\beta_1) + p\mathbb{E}(\beta_2), \quad (15)$$

where $\beta_1 = \min\{t_b - T_D, 0\}$ is the underused time budget, and $\beta_2 = \max\{T_D - t_b, 0\}$ is the total overtime. The objective is to find an optimal time budget allocation t^* that minimizes the time deviation cost in Eq. (15), which is given by:

$$t^* = \arg \min_{t_b \in \Omega} u(t_b). \quad (16)$$

Lemma 1. Let $\beta_1 = \min\{t_b - T_D, 0\}$ and $\beta_2 = \max\{T_D - t_b, 0\}$. Then, it follows that:

$$\mathbb{E}(\beta_1) = \int_0^{t_b} F_{T_D}(t)dt, \quad \mathbb{E}(\beta_2) = \int_{t_b}^{\infty} (1 - F_{T_D}(t))dt \quad (17)$$

Proof. We will start by proving the result for β_1 . Since $\beta_1 \geq 0$, it follows that:

$$\mathbb{E}(\beta_1) = \int_{t=0}^{\infty} P(\beta_1 > t)dt = \int_{t=0}^{t_b} P(t_b - T_D > t)dt = \int_{t=0}^{t_b} P(T_D < t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t)dt.$$

Since β_2 is also non-negative, we can write:

$$\mathbb{E}(\beta_2) = \int_0^{\infty} P(T_D - t_b > t)dt = \int_0^{\infty} P(T_D > t + t_b)dt = \int_{t_b}^{\infty} P(T_D > t)dt = \int_{t_b}^{\infty} (1 - F_{T_D}(t))dt$$

□

Theorem 1. Let $u(t_b)$ be defined in Eq. (15), and t^* be a solution to Eq. (16). Then, it follows that:

$$t^* = F_{T_D}^{-1} \left(\frac{\frac{p}{c}}{1 + \frac{p}{c}} \right), \quad (18)$$

where $F_{T_D}^{-1}(\cdot)$ is the inverse of the cumulative distribution function $F_{T_D}(\cdot)$.

Proof. Substituting (17) in Eq. (15) yields:

$$u(t_b) = c \int_0^{t_b} F_{T_D}(t) dt + p \int_{t_b}^{\infty} (1 - F_{T_D}(t)) dt.$$

When $t_b = t^*$, the first order optimality condition for $u(t_b)$ implies:

$$\frac{du(t_b)}{dt_b} = cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) = 0.$$

Hence, we must have:

$$cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) \implies \frac{F_{T_D}(t^*)}{(1 - F_{T_D}(t^*))} = \frac{p}{c} \implies F_{T_D}(t^*) \left(1 + \frac{p}{c} \right) = \frac{p}{c}$$

We conclude the proof by noting that (18) follows from the expression above. □

Theorem 1 implies that, whenever the remaining time in a session exceeds the quantile t^* in Eq. (18), the surgery is greenlighted to continue. Otherwise, the surgery is cancelled and the session comes to a close. Note that the quantile is solely a function of the ratio between the overtime cost p and the cost of unused session time c , and is applicable to any surgery time distribution. For example, if $p = 2c$, $t^* = F_{T_D}^{-1} \left(\frac{2}{3} \right)$. This means that at least two thirds of the green-lighted surgeries will not require overtime.

4.2.2. Applying the newsvendor model to the case study

Building on the theoretical results of the previous subsection, we first estimate, for each surgical subspecialty, the distribution of the number of surgeries that can be completed within a block of length $T \in \mathbb{Z}_+$. This distribution is then used in the subsequent subsection to evaluate session-planning decisions under an inventory-based formulation. For this purpose, we use the notation introduced in Table 5.

For each subspecialty, let d_i denote the non-negative random variable representing the duration of the i -th surgery, with probability mass function f_d and cumulative distribution function F_d . We assume that surgery durations are independent and identically distributed within each subspecialty. Hence, for any $k \in \mathbb{N}_+$, the cumulative duration of the first k surgeries is

$$cd_k = \sum_{i=1}^k d_i.$$

Let f_{cd_k} and F_{cd_k} denote the probability mass function and cumulative distribution function of cd_k , respectively.

After k surgeries have been completed, one further surgery can be accommodated only if the residual time in the block exceeds the safety threshold t^* . herefore,

$$P(K > k) = P(T - cd_k > t^*) = P(cd_k < T - t^*). \quad (19)$$

Thus, the event $\{K > k\}$ is equivalent to the event that, after completing k surgeries, sufficient time remains in the block to start one additional surgery while preserving the required safety buffer.

Table 6: Symbols used in the derivation of the completion distribution.

Set / Index / Parameter	Description
$\mathcal{S}$	Set of surgical subspecialties
s	Index of a surgical subspecialty, $s \in \mathcal{S}$
T	Length of the operating block
t^*	Safety time threshold that must remain available in the block to accommodate one further surgery
$\varepsilon_{\text{stop}}$	Probability threshold used to truncate the tail of the distribution
K_{max}	Largest value of k such that $P(K > k) > \varepsilon_{\text{stop}}$
Random Variable	Description
d_i	Duration of the i -th surgery for a given subspecialty
cd_k	Cumulative duration of the first k surgeries, $cd_k = \sum_{i=1}^k d_i$
K	Number of surgeries that can be completed within a block
Function / Derived Quantity	Description
f_d	Probability mass function of the duration of a single surgery
F_d	Cumulative distribution function of the duration of a single surgery
f_{cd_k}	Probability mass function of cd_k
F_{cd_k}	Cumulative distribution function of cd_k
f_K	Probability mass function of K
F_K	Cumulative distribution function of K
$*$	Convolution operator
q	Vector with entries $q[k] = P(K > k)$

Because cd_k is the sum of k independent surgery durations, its distribution can be obtained recursively by convolution:

$$f_{cd_1} = f_d, \quad f_{cd_k} = f_{cd_{k-1}} * f_d, \quad k \geq 2.$$

For each k , Eq. (19) then yields the survival probability $P(K > k)$.

These survival probabilities are computed sequentially in Algorithm 1 and stored in the vector q , whose k -th entry is defined as

$$q[k] = P(K > k).$$

The procedure starts at $k = 0$, with $q[0] = 1$, and stops when $q[k] \leq \varepsilon_{\text{stop}}$. This truncation preserves the relevant support of the distribution while avoiding unnecessary convolution steps in the negligible tail. For each subspecialty, the algorithm therefore returns both the vector q and the largest relevant index k , defined as K_{max} .

Algorithm 1 Computation of $P(K > k)$ via convolutions

```

1: Input:  $T, t^*, \varepsilon_{\text{stop}}$ , subspecialties  $s \in \mathcal{S}$ , and duration distributions  $f_d$  for each  $s \in \mathcal{S}$ 
2: Output: vector  $q$  such that  $q[k] = P(K > k)$ 
3: for each  $s \in \mathcal{S}$  do
4:   Initialise  $q[0] \leftarrow 1$ 
5:   Initialise  $k \leftarrow 1$ 
6:   Initialise  $f_{cd_1} \leftarrow f_d$ 
7:   while  $q[k-1] > \varepsilon_{\text{stop}}$  do
8:     if  $k > 1$  then
9:        $f_{cd_k} \leftarrow f_{cd_{k-1}} * f_d$ 
10:    end if
11:     $q[k] \leftarrow P(cd_k < T - t^*)$ 
12:     $k \leftarrow k + 1$ 
13:  end while
14:   $K_{\text{max}} \leftarrow k - 1$ 
15: end for
```

Once the survival function has been obtained, the probability mass function of K follows from the standard relationship between the PMF and the survival function of a discrete random variable:

$$P(K = k) = P(K > k - 1) - P(K > k), \quad k \geq 1. \quad (20)$$

In addition,

$$P(K = 0) = 1 - P(K > 0).$$

Hence, the convolution procedure yields the full probability distribution of the number of surgeries that can be completed within the block for each subspecialty. This distribution is the main output of the present subsection and constitutes the probabilistic input to the inventory-based planning model described next.

4.2.3. Inventory-based model for defining the number of surgeries in a time block

TODO: ADD CITATIONS ABOUT INVENTORY MODEL USED FOR MODELING HERE

Using the distribution of K derived in the previous subsection, we now determine the number of surgeries to be planned within a single operating theatre block. This decision is formulated through an inventory-based perspective that balances the marginal benefit of scheduling additional surgeries against the expected cost of cancellations. The symbols used in this subsection are defined in Table 7.

Table 7: Symbols used in the inventory-based planning model.

Parameter / Decision Variable	Description
n	Number of surgeries planned in a block
n^*	Optimal number of surgeries planned in a block
ℓ	Number of cancelled surgeries
g	Marginal value associated with scheduling one additional surgery
Function / Derived Quantity	Description
$G(n)$	Marginal gain associated with planning n surgeries
$cost(\ell)$	Cost incurred when ℓ surgeries are cancelled
$C(n)$	Expected cancellation cost when n surgeries are planned
$\Delta(n)$	Net value of planning n surgeries
$P(\text{cancellations} = \ell \mid n)$	Probability of cancelling ℓ surgeries when n surgeries are planned

Let $n \in \mathbb{N}_+$ denote the number of surgeries planned in a block. If the random number of surgeries that can actually be completed is K , then the number of cancellations is given by

$$\ell = (n - K)^+,$$

where $(x)^+ = \max\{x, 0\}$. Hence, for $\ell = 0, 1, \dots, n$,

$$P(\text{cancellations} = \ell) = P(K = n - \ell), \quad \ell = 1, \dots, n, \quad (21)$$

and

$$P(\text{cancellations} = 0) = P(K \geq n).$$

We next associate each planning decision with a marginal gain and an expected cancellation cost. In contrast to classical inventory formulations, in which the reward is usually proportional to the expected number of units served, we define the gain of adding an n -th surgery as the value of successfully accommodating the n -th scheduled surgery:

$$G(n) = g \cdot P(K \geq n), \quad (22)$$

where $g > 0$ is a constant representing the marginal value of scheduling one additional surgery. Since

$$P(K \geq n) = P(K > n - 1),$$

Eq. (22) is directly obtained from the survival probabilities computed in the previous subsection.

In this setting, $G(n)$ is not interpreted as a direct monetary return, but rather as the marginal benefit associated with increasing the number of planned surgeries from $n - 1$ to n . This interpretation is particularly suitable in public healthcare settings, where the value of additional surgeries may reflect improved access to care, reduced waiting times, and better utilisation of operating theatre capacity.

The cost associated with cancellations is modelled through a function $cost : \mathbb{Z}_+ \rightarrow \mathbb{R}_+$ satisfying $cost(0) = 0$ and assumed to be non-decreasing. The expected cancellation cost under planning decision n is therefore

$$C(n) = \mathbb{E}[cost((n - K)^+)] = \sum_{x=0}^{\infty} P(K = x) cost((n - x)^+). \quad (23)$$

The net value associated with planning n surgeries is then defined as

$$\Delta(n) = G(n) - C(n). \quad (24)$$

Because $G(n)$ represents the marginal value of adding the n -th surgery, the optimal planning level is the largest value for which the marginal net value remains positive:

$$n^* = \max\{n \in \mathbb{N}_+ : \Delta(n) > 0\}. \quad (25)$$

Thus, n^* identifies the point at which the expected benefit of adding one more planned surgery is no longer sufficient to offset the corresponding expected cancellation cost.

Once n^* has been determined, the relevant probabilistic input for the subsequent operating theatre allocation model is the truncated distribution of the number of surgeries completed under that planning decision, i.e.,

$$P(K = k), \quad k = 0, 1, \dots, n^*,$$

together with the implied probability distribution of the number of cancellations. These quantities are then passed to the next stage of the framework.

4.3. Operating Theatre Scheduling Problem

Following the proposed integrated framework, the final step consists in solving the Operating Theatre Scheduling Problem (OTSP), which addresses the tactical allocation of surgical subspecialties to operating theatre blocks. This stage translates the strategic decisions derived in the preceding modules into implementable schedules for hospital operations. In particular, the OTSP takes as input the target number of surgeries for each sub-speciality, as determined by the (R, Q) policy, together with the effective block capacity obtained from the cancellation and overtime management module.

We formulate the OTSP as a Mixed-Integer Linear Programming (MILP) model. This formulation allows the explicit representation of operational constraints and multiple, potentially competing objectives, while ensuring optimality with respect to the specified criteria. The model is designed to reflect hospital priorities such as high operating theatre utilisation, the allocation of priority subspecialties, and the efficient sequencing of blocks from a logistical perspective.

(Talvez esse parágrafo a seguir fique melhor na seção de Conclusão) The proposed framework is, however, not restricted to the MILP approach adopted here. Since the previous modules provide demand and effective capacity as structured inputs, alternative solution methods, including heuristics and metaheuristics, could also be employed in the OTSP stage. The main value of the framework therefore lies in the integration of decision layers, rather than in the exclusive use of a particular scheduling technique.

In the following subsections, we first describe how the strategic outputs of the (R, Q) policy and the effective capacity information are preprocessed into block-demand units, which constitute the relevant unit of the scheduling problem. We then present the MILP formulation of the OTSP, including its decision variables, objective functions, and constraints.

4.3.1. Preprocessing Strategic Input

The OTSP is formulated at the block level. Therefore, the strategic outputs of the previous modules must be converted into the number of blocks required for each subspecialty.

For each subspecialty $s \in \mathcal{S}$, let

$$p_s(k) = P(K = k), \quad k = 0, 1, \dots, n_s^*,$$

denote the probability mass function of the number of surgeries that can be completed in a single block under the optimal planning decision obtained in the cancellation and overtime management module. This distribution summarises the effective stochastic capacity of one block for subspecialty s .

To determine the number of blocks required to achieve activity targets $\underline{Q}$ and $\overline{Q}$ at confidence level α , we use successive convolutions of $p_s(\cdot)$. Let

$$p_s^{(n)} = \underbrace{p_s * \dots * p_s}_{n \text{ times}}$$

denote the probability mass function of the total number of surgeries that can be completed across n blocks allocated to subspecialty s , and let $F_s^{(n)}$ be its corresponding cumulative distribution function.

Then, for a generic target $Q' \in \{\underline{Q}, \overline{Q}\}$, the minimum number of blocks required is the smallest integer $n \in \mathbb{N}_+$ such that

$$P\left(\sum_{b=1}^n K_b \geq Q'\right) = 1 - F_s^{(n)}(Q' - 1) \geq \alpha, \quad (26)$$

where K_b denotes the number of surgeries completed in block b , assumed independent and identically distributed according to $p_s(\cdot)$.

In other words, we seek the smallest number of blocks such that the probability of completing at least Q' surgeries is no smaller than α . This procedure is applied separately to the lower and upper activity targets. Algorithm 2 summarises the computation.

Algorithm 2 Block demand estimation via successive convolutions

```

1: Input: set of subspecialties  $\mathcal{S}$ , targets  $(\underline{Q}, \overline{Q})$ , confidence level  $\alpha$ , PMF  $p_s$  for each  $s \in \mathcal{S}$ 
2: Output: block requirements  $(\underline{B}_s, \overline{B}_s)$  for each  $s \in \mathcal{S}$ 
3: for each subspecialty  $s \in \mathcal{S}$  do
4:   if  $\underline{Q} = 0$  and  $\overline{Q} = 0$  then
5:      $\underline{B}_s \leftarrow 0, \overline{B}_s \leftarrow 0$ 
6:   else
7:     for each  $Q' \in \{\underline{Q}, \overline{Q}\}$  do
8:        $n \leftarrow 1$ 
9:        $p_s^{(n)} \leftarrow p_s$ 
10:       $F_s^{(n)} \leftarrow \text{CalculateCDF}(p_s^{(n)})$ 
11:      while  $1 - F_s^{(n)}(Q' - 1) < \alpha$  do
12:         $p_s^{(n+1)} \leftarrow \text{Convolve}(p_s^{(n)}, p_s)$ 
13:         $n \leftarrow n + 1$ 
14:         $F_s^{(n)} \leftarrow \text{CalculateCDF}(p_s^{(n)})$ 
15:      end while
16:      if  $Q' = \underline{Q}$  then
17:         $\underline{B}_s \leftarrow n$ 
18:      else
19:         $\overline{B}_s \leftarrow n$ 
20:      end if
21:    end for
22:  end if
23: end for

```

This preprocessing step is essential because the OTSP is defined over block allocations rather than individual surgeries. It also provides a mechanism for propagating the uncertainty captured in the strategic layer into the tactical scheduling stage without explicitly formulating the OTSP as a stochastic optimisation model. As a result, the scheduling problem can be solved as a deterministic MILP while remaining consistent with surgery allocation targets that reflect both variability in surgery durations and the effective capacity of each block.

4.3.2. MILP Formulation

Modeling the OTSP as a MILP allows us to capture the complex interactions between subspecialty demand, block capacity, and scheduling constraints in a structured manner. To formulate the problem, we use the notation in Table 8.

The Objective Functions (OFs) are organised according to the priority levels established by the hospital's guidelines – although they can be reordered as needed to reflect different institutional priorities in other settings. To reflect these priorities, we adopt a hierarchical optimisation strategy, as proposed by Aringhieri et al. (2022). The OFs are optimised sequentially, while respecting the optimal values obtained in the previous stages.

Formally, the first OF is defined as

$$\text{Min } F_0 = \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} z_{s,b} + \gamma \cdot a_b, \quad (27)$$

Table 8: Table of Symbols

Set	Description
OT	Operating theatres (both elective and outpatient)
$\mathcal{D}$	Days of the planning horizon, such that $\mathcal{D} = \{1, 2, \dots, \mathcal{D} \}$
$\mathcal{M}$	Periods (morning or afternoon), such that $\mathcal{M} = \{\text{morning}, \text{afternoon}\}$
$\mathcal{B}$	Blocks, such that $\mathcal{B} = \mathcal{D} \times \mathcal{M}$
$\mathcal{S}$	Surgical subspecialties
$\mathcal{S}_{seq}$	Surgical subspecialties that necessarily require two sequential allocated blocks, such that $\mathcal{S}_{seq} \subset \mathcal{S}$
$\mathcal{S}_{prio}$	but have high priority, such that $\mathcal{S}_{circ} \subset \mathcal{S}$
$\mathcal{S}_{tx}$	Priority surgical subspecialties, such that $\mathcal{S}_{prio} \subset \mathcal{S}$
$\mathcal{S}_{tx}$	Organ transplant surgical subspecialties, such that $\mathcal{S}_{tx} \subset \mathcal{S}$
Index	Description
ot	Operating theatre, with $ot \in OT$
d	Planning horizon day, with $d \in \mathcal{D}$
m	Period (morning or afternoon), with $m \in \mathcal{M}$
b	Surgery block, with $b \in \mathcal{B}$
s	Surgical subspecialty, with $s \in \mathcal{S}$
Parameter	Description
$\bar{B}_s$	Maximum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
B_s	Minimum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
$NumTeam_{b,s}$	Number of teams of subspecialty $s \in \mathcal{S}$ available in block $b \in \mathcal{B}$
$NumAnest_b$	Number of anesthesiologists available in block $b \in \mathcal{B}$
$NeedAnest_s$	1 if subspecialty $s \in \mathcal{S}$ requires an anesthesiologist; 0 otherwise
γ	Constant to penalize the anesthesiologist slack variable a_b in the objective function ($\gamma = \mathcal{B} \cdot \mathcal{S} \cdot OT $)
$\lambda_{ot,b,s}$	1 if room $ot \in OT$, in block $b \in \mathcal{B}$, can be used by subspecialty $s \in \mathcal{S}$ in the shift block $b \in \mathcal{B}$; 0 otherwise
Variable	Description
$x_{ot,b,s}$	1 if room $ot \in OT$, in block $b \in \mathcal{B}$, is allocated to subspecialty $s \in \mathcal{S}$; 0 otherwise. This variable is only created if the allocation is feasible (i.e., specialty s is compatible with the theatre ot).
$y_{ot,d,s}$	1 when room $ot \in OT$, on day $d \in \mathcal{D}$, is allocated only to subspecialty $s \in \mathcal{S}$; 0 otherwise
$w_{b,s}$	Number of transplants to be performed in block $b \in \mathcal{B}$ by subspecialty $s \in \mathcal{S}$
a_b	Slack variable for the number of anesthesiologists available in block $b \in \mathcal{B}$
$z_{s,b}$	Slack variable for the number of teams available in block $b \in \mathcal{B}$ for subspecialty $s \in \mathcal{S}$

which minimises the total slack in team and anesthesiologist availability, with a high penalty on the latter as they are more scarce and expensive. The second function is given by

$$\text{Min } F_1 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{ot,b,s} \text{ only for variables where } \lambda_{otbs} = 0, \quad (28)$$

which minimises the number of allocations outside the aspirational theatre allocation defined for each subspecialty, thus promoting the use of preferred theatres. The third objective function is defined as

$$\text{Max } F_2 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}_{prio}} x_{ot,b,s}, \quad (29)$$

which maximises the allocation of priority subspecialties, ensuring that they receive as much capacity as possible within the constraints of the system. The fourth OF is given by

$$\text{Max } F_3 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{ot,b,s}, \quad (30)$$

which maximises the total number of allocated blocks, thus promoting overall capacity utilisation and service provision. Finally, the fifth OF is defined as

$$\text{Max } F_4 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{ot,b,s}, \quad (31)$$

which maximises the number of sequentially allocated blocks. This is particularly important for improving the logistics of the surgical process, reducing the need for equipment and team transfers between theatres.

With this, we formulate the problem, incorporating these OFs and constraints, but with the flexibility to accommodate other settings as needed.

(*OTScheduling*) Optimize F_0 – F_5

subject to

$$\sum_{ot \in OT} \sum_{b \in \mathcal{B}} x_{ot,b,s} \leq \bar{B}_s \quad \forall s \in \mathcal{S} \quad (32)$$

$$\sum_{ot \in OT} \sum_{b \in \mathcal{B}} x_{ot,b,s} \geq \underline{B}_s \quad \forall s \in \mathcal{S} \quad (33)$$

$$\sum_{s \in \mathcal{S}} x_{ot,b,s} \leq 1 \quad \forall ot \in OT, b \in \mathcal{B} \quad (34)$$

$$\sum_{ot \in OT} \sum_{s \in \mathcal{S}} NeedAnest_s \cdot x_{ot,b,s} \leq NumAnest_b + a_b \quad \forall b \in \mathcal{B} \quad (35)$$

$$\sum_{ot \in OT} x_{ot,b,s} \leq NumTeam_{bs} + z_{s,b} \quad \forall b \in \mathcal{B}, s \in \mathcal{S} \quad (36)$$

$$x_{ot,b,s} + x_{ot(b+1)s} - 1 \leq y_{ot,d,s} \quad \forall ot \in OT, b \in \mathcal{B}, s \in \mathcal{S}, d \in \mathcal{D} : b \text{ and } b+1 \text{ are blocks of the same day } d \quad (37)$$

$$x_{ot,b,s} + x_{ot(b+1)s} \geq 2 \cdot y_{ot,d,s} \quad \forall ot \in OT, b \in \mathcal{B}, s \in \mathcal{S}, d \in \mathcal{D} : b \text{ and } b+1 \text{ are blocks of the same day } d \quad (38)$$

$$x_{ot,b,s} = x_{ot(b+1)s} \quad \forall ot \in OT, b \in \mathcal{B}', s \in \mathcal{S}_{seq}, d \in \mathcal{D} : b \text{ and } b+1 \text{ are blocks of the same day } d \quad (39)$$

$$\sum_{ot \in OT} x_{ot,b,s} = 2 \cdot w_{b,s} \quad \forall b \in \mathcal{B}, s \in \mathcal{S}_{tx} \quad (40)$$

$$x_{ot,b,s} \in \{0, 1\} \quad \forall ot \in OT, b \in \mathcal{B}, s \in \mathcal{S}$$

$$y_{ot,d,s} \in \{0, 1\} \quad \forall ot \in OT, d \in \mathcal{D}, s \in \mathcal{S}$$

$$w_{b,s} \in \mathbb{Z}_+ \quad \forall b \in \mathcal{B}, s \in \mathcal{S}$$

$$a_b \in \mathbb{Z}_+ \quad \forall b \in \mathcal{B}$$

$$z_{s,b} \in \mathbb{Z}_+ \quad \forall b \in \mathcal{B}, s \in \mathcal{S}$$

Constraints (32) and (33) impose minimum and maximum bounds on the number of allocations of each subspecialty. This is part of the integration with the strategic layer, as these bounds are integrated to (R, Q) policy and designed to enforce a prescribed waiting time requirement subspecialty, while taking full consideration of the uncertainty in surgery durations and the associated risk of surgery cancellations. Constraint (34) ensures that each theatre and time block receives at most one subspecialty. Constraint (35) is a soft constraint that limits the number of allocations that require anaesthetists according to their availability $NumAnest_b$ in each block, while allowing for some flexibility through the slack variable a_b . Constraint (36) similarly limits the number of allocations for each subspecialty based on the availability of medical teams $NumTeam_{bs}$, with the slack variable $z_{s,b}$ providing flexibility when necessary. Constraints (37) and (38) together guarantee sequential allocation of blocks on the same day when $y_{ot,d,s} = 1$. Subspecialties requiring two consecutive blocks are specifically addressed by constraint (39). Constraint (40) ensures that transplant subspecialties are allocated in pairs, reflecting the need for simultaneous operating theatre use for donor and recipient procedures. tion balancing in conjunction with OF (29).

No explicit constraints are required to model theatre-subspecialty compatibility. Allocation variables $x_{ot,b,s}$ are introduced only for compatible pairs (s, ot) , where theatre $ot \in OT$ provides the infrastructure required by subspe-

cialty $s \in \mathcal{S}$. Hence, infeasible assignments are excluded a priori from the feasible region, and all feasible solutions automatically satisfy compatibility requirements.

5. Computational Study

We conduct a computational study to evaluate the performance of the proposed approach using real-world data from HC - Unicamp. While the theoretical results provide guarantees on system stability, it is essential to assess the practical implications in a realistic setting. To this end, we simulate the long-term surgery planning process and compare the proposed approach with a baseline policy in terms of patient waiting times and resource utilisation.

5.1. Real-world Data

The dataset comprises detailed information on surgery durations, operating theatre availability, team constraints, and specialty schedules over a one-year period (2025). Each month is treated as an individual instance of the OTSP, while the full year is used to evaluate long-term system behaviour.

5.2. Derivation of Model Inputs

The data are processed to derive the inputs required for the strategic and operational components of the proposed approach.

Demand distributions. Patient arrivals are modelled as Poisson processes. For each subspecialty, the arrival rate λ is estimated as the average monthly demand derived from historical activity. This parameter is used in the Queue Management module to determine the (R, Q) policy parameters, ensuring that waiting-time requirements are met with high probability despite demand uncertainty.

Surgery duration distributions. For each subspecialty, empirical distributions of surgery durations are estimated from historical data. These distributions are used in the Overtime and Cancellation module to evaluate the newsvendor model and determine the number of surgeries scheduled per block; see Section 4.2.

Operational parameters. All parameters of the operating theatre scheduling model (see Table 8 in Section 4.3) are based on real-world data, i.e., the actual values observed in each month of the year 2025. These include operating theatre compatibilities, staff availability, and working calendars.

5.3. Experimental Setup

Experiments are conducted on a machine equipped with an Intel Core 7 150U processor (5.40 GHz) and 32 GB of RAM. All modules are implemented in Python 3.12, and the MILP model is solved using SCIP 9.2.3 with default parameter settings, interfaced through the Pyomo library.

For the Queue Management module, the waiting time target is set to $\tau = 6$ months with confidence level $\alpha = 0.1$, ensuring that patients are treated within this horizon with high probability. The threshold R is defined as the midpoint between Q and $\bar{Q}$. In the Overtime and Cancellation module, the overtime cost is set to twice the regular surgery cost, while the cancellation cost equals the regular cost. Additionally, we set the cost function to be quadratic in the number of surgeries cancelled, i.e., $c(x) = x^2$, reflecting the increasing marginal cost of cancellations as more surgeries are cancelled. These strategic parameters (modules 1 and 2) are computed once and remain fixed throughout the simulation horizon.

Baseline policy. The baseline policy excludes the strategic modules. All specialties are scheduled in every period, and the MILP model is solved using only operational constraints, without (R, Q) control or overtime optimisation. Essentially, we do so by deactivating the block demand constraints (32) and (33) in the MILP model and using the original surgery duration distributions (without adjustment for overtime or cancellations) to draw the number of surgeries that happened into each scheduled block. The patient arrival process and the operational parameters remain the same as in the proposed approach, ensuring a fair comparison.

5.4. Simulation Framework

Figure 4 illustrates the simulation flow. The system evolves over a monthly decision cycle. In each period, patient arrivals are realised according to the estimated demand distributions, and queues are updated accordingly. Under the proposed approach, only specialties with queue size exceeding R are selected for scheduling.

The MILP model then determines the allocation of operating theatre blocks. For the proposed approach, the number of surgeries is sampled based on the distributions obtained from the Overtime and Cancellation module. Queues are updated after the month cycle, and the process repeats at each new period.

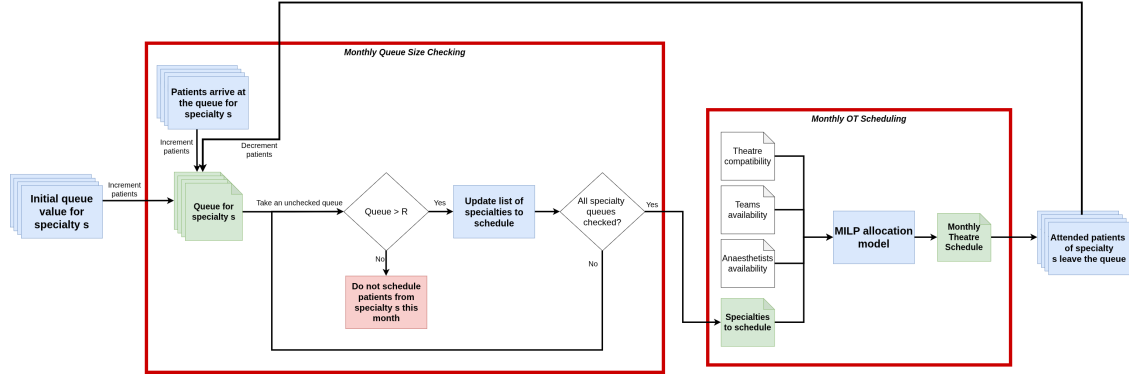


Figure 4: Long-term simulation flow.

To assess long-term stability of the framework and the baseline, we perform 100 independent simulation runs each, using different random seeds to capture the stochasticity in surgery durations and patient arrivals. This way, we can evaluate not only the expected performance but also the variability and robustness of the proposed approach compared to the baseline policy.

5.5. Results from Strategic Decision-Making

5.5.1. (R, Q) Batch Size Results

Figure 5 summarises the control intervals obtained by the (R, Q) policy across all subspecialties. To improve readability, subspecialties are ordered according to the interval width, $\bar{Q} - \underline{Q}$, rather than alphabetically. The figure should therefore be interpreted as an ordered profile of the control bands generated by the policy, while the complete set of limits is reported in Appendix ?? Definir Apendice.

The results reveal substantial heterogeneity across subspecialties. Although most services are associated with relatively narrow control bands, a smaller group exhibits considerably higher lower and upper limits. This pattern is consistent with the patient demand distributions and the (R, Q) policy itself. Subspecialties with larger expected demand tend to receive higher values of both $\underline{Q}$ and $\bar{Q}$, whereas services with greater demand variability require wider admissible replenishment ranges.

Table 9 illustrates how this heterogeneity translates into operational release limits. The high-interval subspecialties, such as ENT (Ear, Nose, Throat), neurosurgery, and vascular surgery, receive substantially larger control bands, indicating that the number of patients released for scheduling has the flexibility to vary considerably from one planning cycle to another. In a high-demand public hospital, this flexibility is important because broad-access or high-referral subspecialties may accumulate patients rapidly when surgical capacity is constrained. A narrow or uniform release rule would therefore risk limiting the candidate mix available to the tactical scheduling model.

By contrast, the low-interval subspecialties operate within very tight control bands. This does not imply lower clinical importance; rather, it indicates that their candidate mix are smaller and more stable from the perspective of queue replenishment. The medium group occupies an intermediate position, representing services whose demand is recurrent enough to require some flexibility, but not sufficiently large or variable to dominate the release process. Thus, the (R, Q) policy avoids imposing a common batch size across the system and instead adapts the admissible replenishment range to the demand profile of each subspecialty.

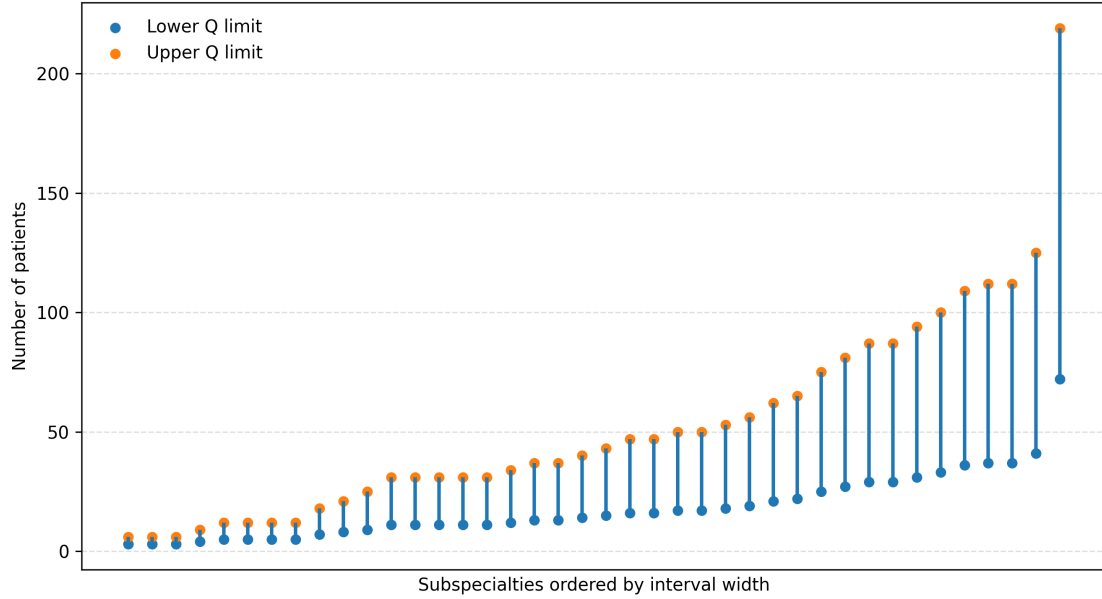


Figure 5: Control intervals for the (R, Q) policy across subspecialties.

These results have direct implications for CMP. As the (R, Q) limits determine the mix of patients transferred from the queue-management layer to the tactical planning stage, subspecialties with wider and higher intervals are more likely to influence capacity allocation and overtime requirements. This does not necessarily mean that these services should always receive more operating-theatre time. In fact, it indicates that their larger and more variable candidate mix must be explicitly represented in the tactical optimisation problem. Conversely, subspecialties with narrow intervals generate smaller and more predictable candidate sets, reducing their impact on capacity allocation decisions.

Additionally, it is important to note that the (R, Q) policy only triggers allocation decisions to a subspecialty when its queue size exceeds R . This means that even subspecialties with wider intervals will not necessarily dominate the block allocation process, as they will only be considered for scheduling when their queues are sufficiently large. This mechanism ensures that the system remains responsive to actual demand conditions, rather than being driven solely by the potential variability in patient arrivals.

Therefore, the control intervals provide a quantitative link between queue dynamics and tactical planning. They

Table 9: Representative subspecialties selected from the lower, middle, and upper parts of the interval-width distribution generated by the (R, Q) policy.

Interval group	Subspecialty	$\underline{Q}$	$\bar{Q}$	$\bar{Q} - \underline{Q}$
High interval	ENT	72	219	147
	Neurosurgery	41	125	84
	Vascular surgery	37	112	75
Medium interval	Spine orthopaedics	15	43	28
	Kidney/gastro	14	40	26
	Knee orthopaedics	13	37	24
Low interval	BMT	3	6	3
	Dermatology	3	6	3
	Shoulder orthopaedics	3	6	3

translate differences in subspecialty-specific demand into structured release limits, ensuring that the scheduling modules receive patient mix that are consistent with both the statistical behaviour of the queues and the operational characteristics of a high-demand public hospital such as HC - Unicamp.

5.5.2. Overtime and Cancellation Results

The (R, Q) results defined the target batch-size intervals required to regulate waiting-list levels across subspecialties. However, these targets must be integrated with block-level surgical plans before they can be used by the OTSP module. The overtime and cancellation module performs this planning by estimating, for each subspecialty, how many surgeries should be planned per block under uncertain surgical durations. This subsection reports the experimental results of this module, focusing on Foot Orthopaedics as a representative subspecialty, although the same analysis is performed for all subspecialties.

Figure 6 shows the probability distribution of the number of surgeries completed in a Foot Orthopaedics block for different planned values of n . The distributions are computed after applying the newsvendor-based cancellation rule defined in Section 4.2.1, according to which an additional surgery is only started if the remaining session time is sufficiently large. Thus, the figure represents the effective surgical output of a block, rather than its nominal planned workload. **O tamanho da fonte desse grafico 6 (o seguinte tbm) ta discrepante com os outros, tem que ser ajustado.**

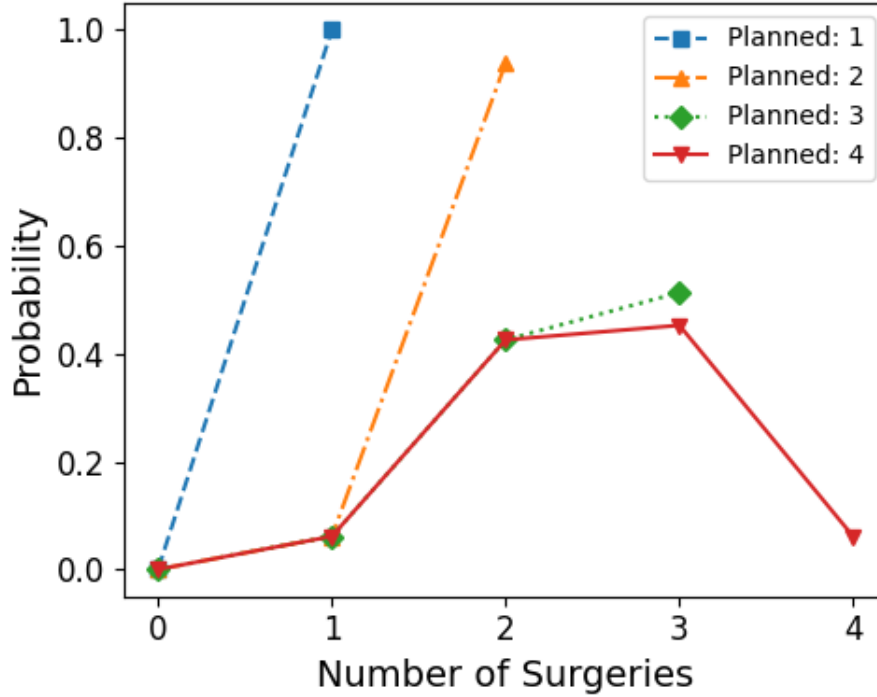


Figure 6: Probability distribution of completed surgeries in Foot Orthopaedics blocks for different planned values of n .

The results show that conservative planning almost eliminates cancellation risk, but at the cost of low block productivity. When $n = 1$, the planned workload is always completed, indicating that one surgery per block is a safe but inefficient decision for this subspecialty. For $n = 2$, the probability mass remains highly concentrated around two completed surgeries, suggesting that the block can usually absorb this workload without substantial cancellation risk. When $n = 3$, the distribution becomes more dispersed: completing all three surgeries remains feasible, but the probability of completing only two surgeries becomes relevant. This indicates that the third planned surgery increases expected throughput while also exposing the system to a non-negligible risk of cancellation. Finally, for $n = 4$, the probability of completing all planned surgeries becomes very low, and the realised output is more likely to remain below the nominal plan.

These results provide an important insight into the trade-off between productivity and cancellation risk. The block does not behave as a deterministic capacity unit capable of accommodating a fixed number of surgeries with certainty. Instead, increasing n changes the probability distribution of realised surgical output. Therefore, using only nominal capacity would overestimate the contribution of a block when many surgeries are assigned to it. This is particularly relevant for the proposed framework, since the (R, Q) policy may indicate the need to treat a certain number of patients in the month cycle, but the overtime module determines how much of this demand can realistically be assigned to each block without creating excessive cancellation risk.

Figure 7 complements this result by comparing the expected marginal gain and the expected cancellation cost associated with each value of n for the same subspecialty. While Figure 6 shows the probability of different realised outputs, Figure 7 converts this uncertainty into a decision criterion for selecting the planned number of surgeries per block, as defined by the inventory-based model in Section 4.2.3.

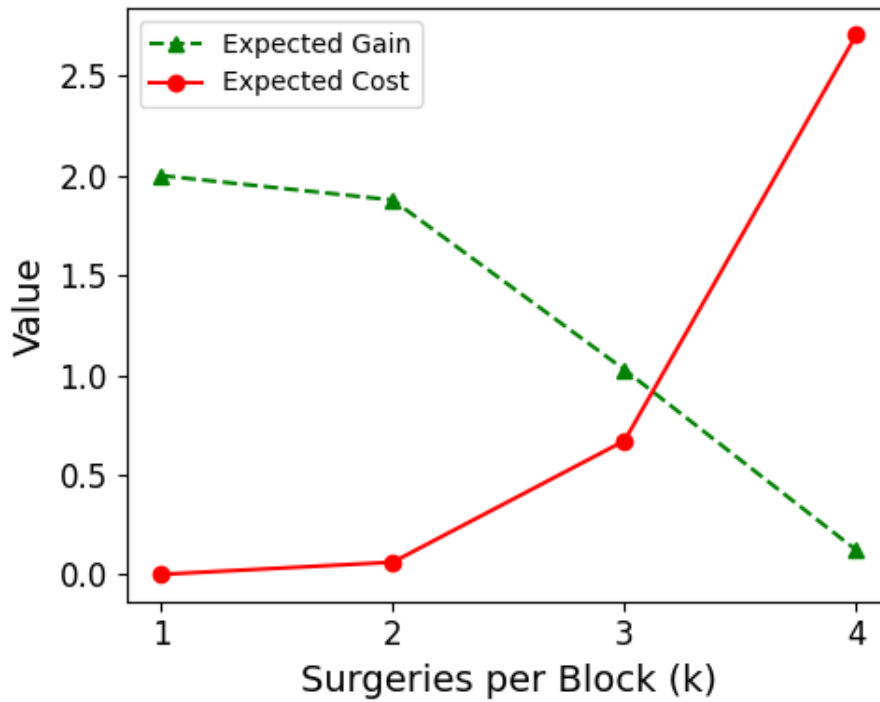


Figure 7: Expected marginal gain and expected cancellation cost associated with planning n surgeries in a Foot Orthopaedics block.

The gain–cost comparison confirms that planning too few surgeries underuses available operating theatre capacity, whereas planning too many surgeries produces an unfavourable cancellation profile. For $n = 1$ and $n = 2$, the expected gain is substantially higher than the expected cost, indicating that these planning levels are operationally safe. However, they are also conservative, especially when compared with the additional throughput that may be obtained by planning a third surgery. For $n = 3$, the expected cancellation cost increases, but it remains below the expected gain. This suggests that the third surgery is still justified, since it improves expected block productivity without making the cancellation risk dominant.

The main change occurs when moving from $n = 3$ to $n = 4$. At this point, the expected gain drops sharply, while the expected cancellation cost increases substantially. This pattern indicates that the fourth planned surgery contributes little to expected completed output and mainly increases the likelihood of cancellations. Accordingly, the experimental result for Foot Orthopaedics supports $n^* = 3$ as the recommended number of surgeries to plan per block. This value represents the highest planning level for which the expected benefit of adding surgeries still compensates for the expected cancellation cost.

Thus, the overtime and cancellation results show that the effective capacity of each block is subspecialty-specific

as it depends on the empirical distribution of surgical durations. For Foot Orthopaedics, as an example, the experiment indicates that three planned surgeries per block provide the best balance between productivity and cancellation risk. This output is then used by the operating theatre scheduling module after being combined with the (R, Q) limits to determine how many blocks should be allocated to this subspecialty in each month. Consequently, the MILP does not allocate blocks based only on nominal session availability, but on a probabilistic estimate of how many surgeries each allocated block can realistically deliver while controlling overtime-related cancellations.

5.6. Long-term Stability Analysis

In this section, we evaluate the long-term stability of the system under the proposed approach and compare it with the baseline policy. We analyse the evolution of patient queues over time and the allocation of operating theatre blocks, highlighting the differences between the two approaches.

5.6.1. Queue Management Results

The one-year simulation horizon enables a detailed characterisation of queue dynamics under both the proposed framework and the baseline policy. We analyse the results obtained from the 100 independent simulation runs and capture the variability the 10–90% percentile interval, allowing us to assess not only expected performance but also robustness under stochastic demand.

Figure 8 presents the evolution of the queue for the subspecialty *Upper Digestive Endoscopy (UDP) – Pediatric*. Under the proposed approach (Figure 8a), the system exhibits two distinct operational regimes. Initially, a transient phase is observed, in which the queue decreases steadily from its initial level. This phase is followed by a controlled regime, where the queue starts to stabilise (approximately between months 6 and 9) and then fluctuates around the threshold R . This transition reflects the action of the (R, Q) policy, which introduces a demand-driven mechanism: allocation decisions are triggered only when the queue exceeds R , thereby preventing unnecessary servicing once the backlog has been reduced. As a result, the system converges towards a stable operating point, with a stabilization in variability across simulation runs, as indicated by percentile band. (Preciso alterar as cores do grafido da evolucao da fila do baseline e colocar fontes maiores.)

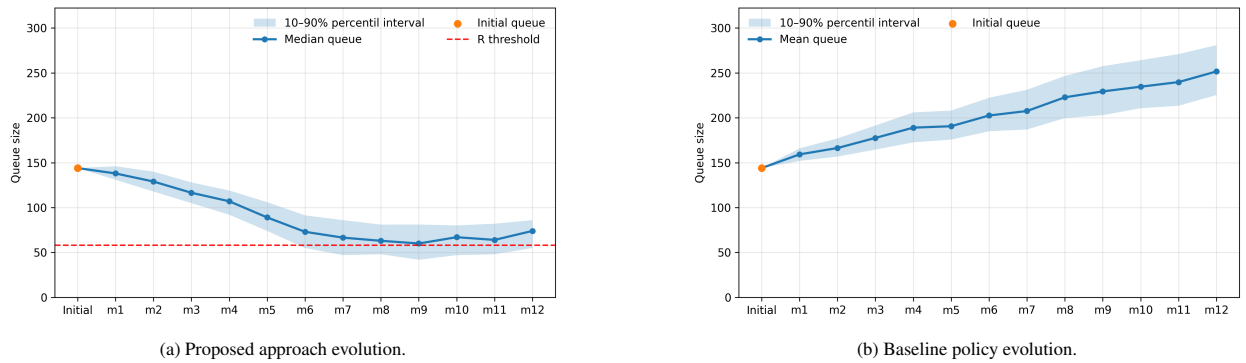
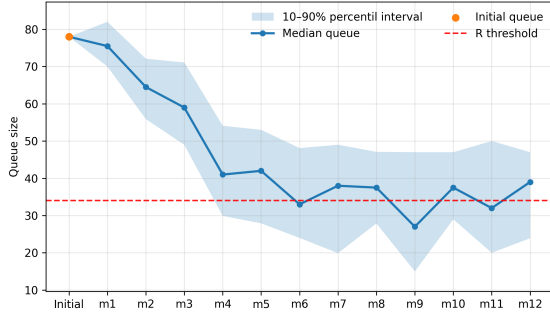


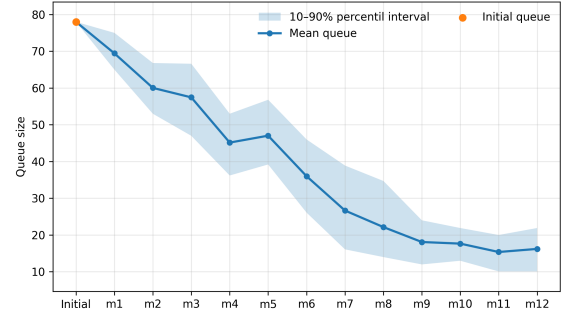
Figure 8: Evolution of patient queues over time for the subspecialty *UDP – Pediatric*.

In contrast, the baseline policy (Figure 8b) operates in a queue-agnostic allocation rule, with allocation decisions independent of the queue state. Although a short-term reduction is observed, the queue subsequently exhibits sustained growth and increasing dispersion. This behaviour indicates a misalignment between demand and capacity allocation, leading to backlog accumulation and reduced predictability. The widening variability further suggests that the system becomes increasingly sensitive to stochastic fluctuations, which may translate into unstable waiting times in practice.

A similar but more pronounced pattern is observed for the subspecialty *Gastroenterology – Esophagus-Stomach-Duodenum (ESD)* in Figure 9. Under the proposed approach (Figure 9a), the initial depletion phase occurs more rapidly, reflecting the higher effective service capacity of this subspecialty. Subsequently, the queue converges to the same controlled regime around R , indicating that the policy adapts dynamically to differences in service rates while maintaining a consistent target operating level across subspecialties. This highlights that the (R, Q) mechanism is not



(a) Proposed approach.



(b) Baseline policy.

Figure 9: Evolution of patient queues over time for the subspecialty *Gastroenterology – ESD*.

merely reactive, but also self-regulating, ensuring both convergence and stability irrespective of capacity heterogeneity.

For the baseline policy (Figure 9b), the queue also decreases over time due to the relatively high allocation frequency of this subspecialty. However, this apparent improvement is local rather than systemic. As the baseline seeks to allocate all subspecialties uniformly, without accounting for their relative queue levels, it fails to prioritise resources where they are most needed. Consequently, capacity may be over-allocated to subspecialties with already manageable queues, while others remain congested. This imbalance is not visible when analysing a single queue in isolation but becomes critical at the system level, where it leads to inefficient resource utilisation and degraded overall performance.

These results have important implications for the integration of the queue management module with the subsequent components of the framework. First, by regulating queues around a target level, the (R, Q) policy provides a stabilised and informative input to the overtime planning module, which determines the number of surgeries to be performed within each block. In particular, the reduced variability and bounded queue sizes lead to more reliable estimates of demand, thereby improving the calibration of safety margins and reducing the risk of either under-utilisation or excessive overtime.

Second, the stabilised queue profiles directly benefit the operating theatre scheduling module. By filtering out extreme fluctuations and preventing backlog accumulation, the queue management module effectively regularises the input to the scheduling problem, reducing the likelihood of infeasible or highly imbalanced allocations. This contributes to improved tractability and solution quality, as the MILP operates on a more balanced and predictable set of requirements across subspecialties.

Therefore, the (R, Q) -based mechanism acts as a coordination layer between stochastic demand and deterministic scheduling decisions. By inducing convergence to a steady-state regime, reducing variability, and enabling demand-driven prioritisation, it enhances both the robustness and the efficiency of the downstream planning modules. In contrast, the baseline policy fails to provide such coordination, which may result in locally acceptable but globally inefficient solutions. Comprehensive queue results for all subspecialties are provided in Appendix ??.

5.6.2. Long-term OT Allocation Results

This subsection analyses the long-term allocation decisions produced by the OT Scheduling module (see Section ??). While the (R, Q) module defines when each subspecialty requires intervention and the overtime module estimates how many surgeries can be planned per allocated block, the OT Scheduling module converts these upstream decisions into a feasible monthly allocation of operating theatre capacity. Therefore, this module acts as the operational interface between queue-control decisions and theatre-level planning.

In the proposed framework, the MILP does not allocate individual patients directly. Instead, it assigns theatre blocks to subspecialties. For each allocated block, the expected number of planned surgeries is obtained from the overtime module, which accounts for the trade-off between overtime and cancellation risk. This modelling structure allows the MILP to operate at the same aggregation level used in practical theatre planning, while preserving the connection with patient-level demand through the queue-control policy.

We analyse the monthly operating room occupancy obtained under the proposed framework and the baseline

policy in Figure 10. The results show that the OT Scheduling module is able to generate feasible allocations with substantial use of available capacity in both approaches. This is consistent with the MILP formulation, since the model allocates blocks while respecting theatre availability, subspecialty compatibility, monthly capacity limits, and the surgical volumes determined by the upstream modules. Thus, the scheduling layer is able to translate the planned surgical demand into executable theatre allocations.

(Na verdade, tanto esse quanto o grafico seguinte podem ser graficos de linha, porque tem uma relacao temporal nisso...)

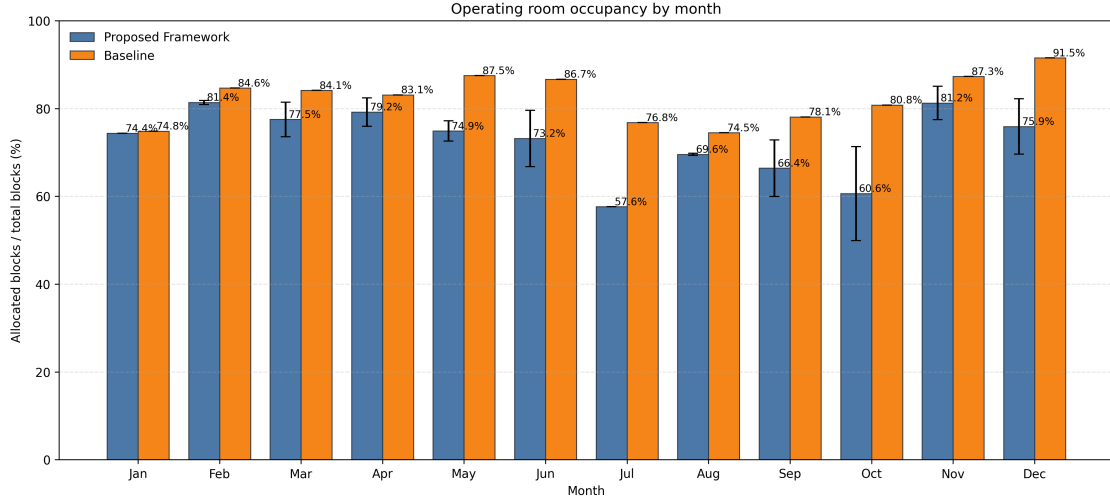


Figure 10: Monthly operating room occupancy under the proposed framework and the baseline policy.

However, the baseline systematically presents higher occupancy than the proposed framework. This difference is expected and reflects the distinct logic behind the two allocation policies. The baseline allocates capacity without explicitly considering whether each subspecialty queue is above or below its reorder point. As a consequence, it tends to distribute theatre blocks across a larger set of subspecialties and maintains high occupancy even when some queues do not require immediate intervention.

By contrast, the proposed framework only triggers allocation for a subspecialty when its queue exceeds the corresponding reorder threshold R . If the queue is below this level, the framework does not force an allocation merely to use available theatre capacity. Therefore, the lower occupancy observed in some months should not be interpreted as a scheduling inefficiency. Rather, it indicates that the framework avoids unnecessary interventions when the queue-control policy identifies that a subspecialty is already within a controlled region. This behaviour is particularly visible in months such as July, May and October, where the proposed framework uses a smaller share of the available capacity than the baseline.

Figure 11 complements this interpretation by showing the number of subspecialties allocated in each month. The baseline allocates a relatively stable and high number of subspecialties throughout the simulation horizon, ranging from approximately 34 to 39 subspecialties per month. In contrast, the proposed framework allocates fewer subspecialties in most months, with a more pronounced reduction in the second half of the year. This result confirms that the difference in occupancy is directly associated with the queue-control mechanism: when fewer subspecialties require intervention, fewer subspecialties are scheduled.

This behaviour has important implications for long-term planning. A policy that keeps allocating blocks to all or nearly all subspecialties may achieve higher occupancy, but it can also consume capacity on queues that are already under control. The proposed framework introduces a more selective allocation logic: capacity is concentrated on subspecialties whose queues require action, while subspecialties below R can temporarily remain unallocated. This creates a controlled form of flexibility, because the MILP can redirect blocks across months according to the evolving state of the system rather than reproducing a fixed allocation pattern.

The role of this flexibility becomes clearer when the realised allocations are compared with the admissible intervals

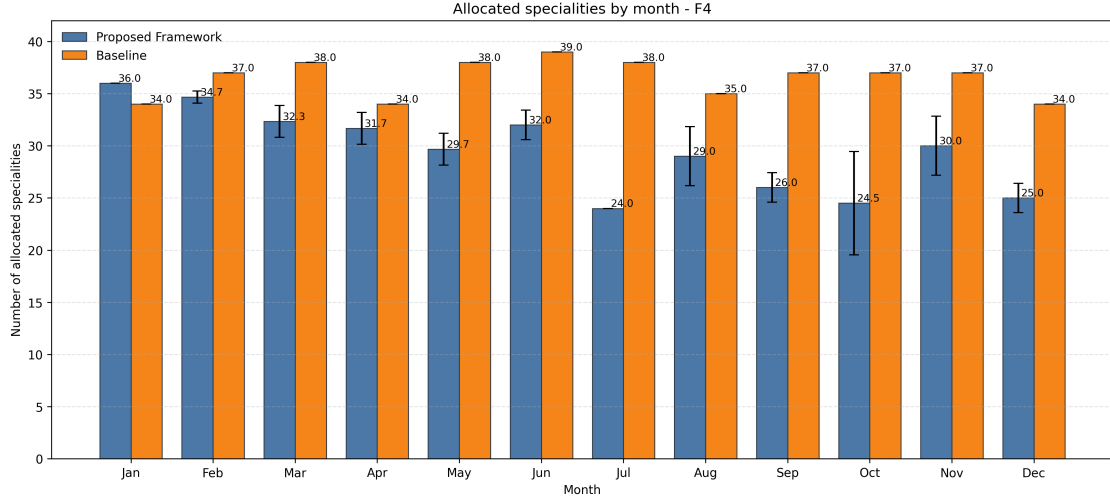


Figure 11: Number of subspecialties allocated per month under the proposed framework and the baseline policy.

generated by the (R, Q) policy. Figures 12 and 13 show, for selected subspecialties, the realised number of allocated surgeries in relation to their corresponding Q intervals at the beginning and at the end of the simulation horizon. The horizontal segments represent the admissible interval $[Q, \bar{Q}]$, the vertical tick indicates the arrival-rate parameter λ , and the shaded regions summarise the realised variability across simulation seeds. The markers indicate whether the median allocation is below, within, or above the corresponding Q interval.

(Percebi que preciso escolher entre: detalhar as implicacoes de aconecer alocaoes abaixo do intervalo e abaixo do lambda, ou só tirar isso dos graficos...)

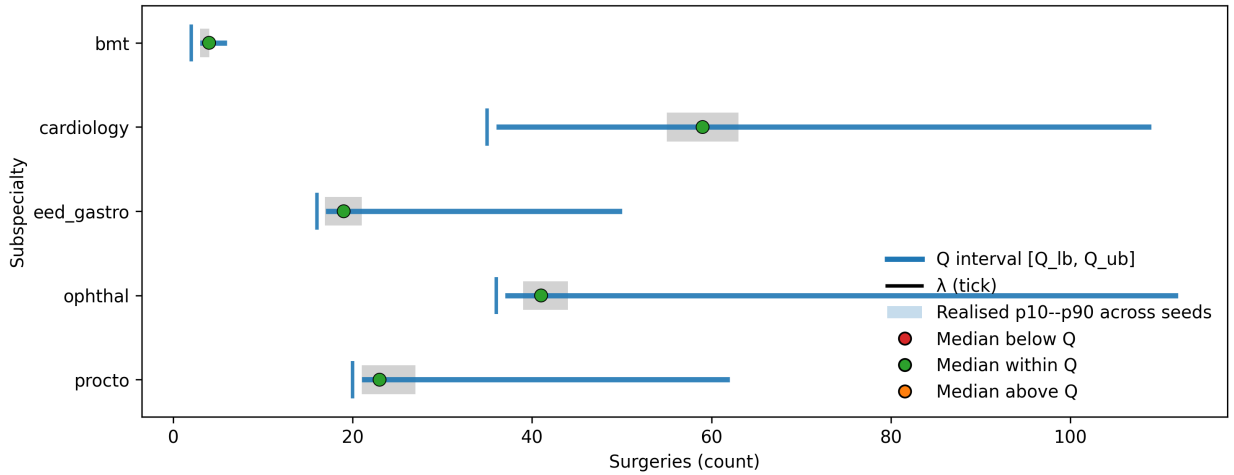


Figure 12: Allocated surgeries relative to the Q intervals in the first month.

The comparison between the first and last months shows that allocation decisions are not fixed over time. In the first month, the median allocation for some subspecialties is positioned close to the lower part of the admissible interval, while others are allocated closer to the centre or above the interval. By the last month, the relative position of the allocations changes: the selected subspecialties present median allocations within their respective Q intervals. This indicates that the framework adjusts theatre allocation as queues evolve, rather than imposing a static monthly allocation target.

This result is especially relevant for subspecialties with wider Q intervals, such as *cardiology*, *ophthal*, *procto*,

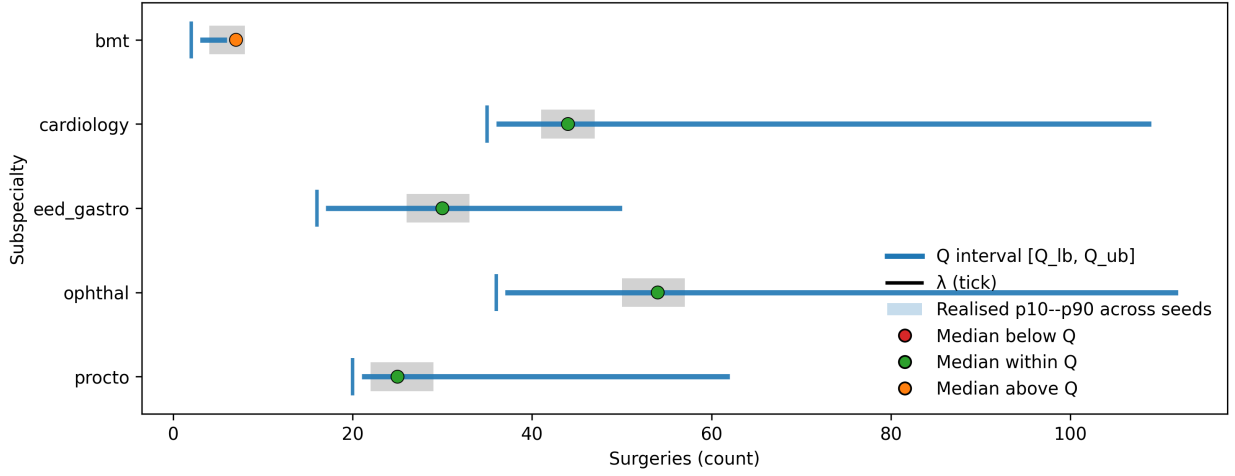


Figure 13: Allocated surgeries relative to the Q intervals in the last month.

and *eed_gastro*. A wider interval reflects greater variability in the monthly arrival distribution and, consequently, a broader admissible range for the number of surgeries that can be allocated when intervention is required. The MILP can exploit this range to accommodate competing priorities across subspecialties while remaining consistent with the queue-control policy. For example, when one subspecialty is below its reorder point, the framework can omit or reduce its allocation and use the available capacity for another subspecialty with greater queue pressure.

This controlled flexibility is a key distinction between the proposed framework and the baseline. The baseline tends to preserve a broader and more regular allocation of subspecialties, which leads to higher occupancy but weaker responsiveness to the actual state of the queues. The proposed framework, instead, links allocation decisions to queue status: it may increase, reduce, or temporarily suppress allocations depending on whether each subspecialty requires intervention. As a result, the framework does not simply aim to fill the operating room schedule; it aims to allocate theatre capacity where it contributes to maintaining queue stability.

Overall, the long-term allocation results show that the OT Scheduling module successfully implements the decisions produced by the queue-control and overtime modules. The occupancy results demonstrate that the MILP can generate feasible and operationally meaningful theatre plans. The lower occupancy observed under the proposed framework is consistent with the (R, Q) policy, since capacity is not forced into subspecialties whose queues are already below the reorder threshold. The reduction in the number of allocated subspecialties confirms this selective behaviour, while the comparison with the Q intervals shows that the framework preserves flexibility within controlled allocation bounds. Therefore, the proposed framework provides a planning mechanism that is not only feasible at the operating theatre level, but also more responsive to the long-term dynamics of surgical queues.

6. Conclusions and Future Directions

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